RESULTS.md — one row per reported number (schema below)

R### | quantity | value | 95% CI | shots x seeds | producing cell/file | timestamp

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R001 | Gate 0 benchmark (`python scripts/gate0_benchmark.py`) | GATE0 PASS (all asserts G0.1-G0.9 true) | n/a | n/a (exact diagonalisation, no shots) | scripts/gate0_benchmark.py, run via .venv (2026-08-12) | 2026-08-12 R002 | Checkpoint 2 (model matches spec) | all 8 asserts PASS: [H,Q]=0.00e+00; 4 populated levels, sectors {+1,+3}; |000> eigenstate at E=0.55; Q=3 weight 0.633749; no aliasing (radius 2.64<15.71); resolvable (spacing 1.03>0.50) | n/a (deterministic/exact) | n/a | notebook cell 16, executed headlessly via nbconvert prefix run (.venv, kernel qh26-t5) | 2026-08-12 R003 | Checkpoint 3a (`build_controlled_evolution`, Challenge 1) | exact path max|c-U - block|=6.87e-16 (PASS, <1e-10); Trotter error reps=1/2/8: 8.22e-01/1.68e-01/9.67e-03, monotonically decreasing (PASS), reps=8 accurate (PASS, <2e-2) | n/a (deterministic operator identity) | n/a | notebook cell 24, executed headlessly via nbconvert prefix run | 2026-08-12 R004 | Checkpoint 3b (sign convention, all 4 conventions at once) | max deviation over 3 times x 2 quadratures x 5 observables = 1.33e-15 (PASS, <1e-10); D1 marginal ||diff||=3.50e-16 (PASS); ancilla-marginal-undisturbed-by-shadow-rotation (PASS) | n/a (deterministic) | n/a | notebook cell 26, executed headlessly via nbconvert prefix run | 2026-08-12 R005 | Checkpoint 3c (measurement order irrelevant, algebra + Aer experiment) | POVMs commute (max 0.0e+00); p(a,s) identical global/anc-first/sys-first (max diff 1.4e-16); Aer TVD to exact: terminal 0.00211, mid-circuit 0.00261 (both < 5x noise floor 0.0036) | n/a | 200,000 shots (this checkpoint only, given code) | notebook cell 29+31, executed headlessly via nbconvert prefix run | 2026-08-12 R006 | Checkpoint 4 (`parse_memory`, `run_shadow_hadamard`, Challenge 3) | worked example exact match: parse_memory(["0100","1011"],n=3) -> ancilla=[1,-1], system= [[1,1,-1],[-1,-1,1]] (PASS, matches CONVENTIONS.md §1 verbatim); fixed-seed reproducibility bit-identical (PASS); basis frequencies [0.336,0.335,0.329] vs 1/3 (PASS); kept records == requested 500 (PASS); n_circuits=27=3^3 for 500 shots (PASS) | n/a (deterministic checks) | 500 shots (repro test) | notebook cell 34 (implementation) + cell 36 (checkpoint), executed headlessly via nbconvert prefix run | 2026-08-12 R007 | Checkpoint 5 (three estimators, Challenge 4, t=0.9) | 9/9 PASS, all <= 1.2 sigma (5 sigma gate; one value, chi_Z0 im, is exactly 1.2 -- C4 audit note): chi re +0.6577 vs +0.6623 (0.5s); chi im -0.1327 vs -0.1320 (0.1s); +0.0953 vs +0.0739 (0.6s); “ +2.2477 vs +2.2675 (0.8s); (rho^(I) ref) +0.5150 vs +0.5258 (0.7s); chi_Q re +1.7855 vs +1.7777 (0.2s); chi_Q im -0.6880 vs -0.7341 (1.1s); chi_Z0 re +0.4725 vs +0.4530 (0.9s); chi_Z0 im -0.4435 vs -0.4701 (1.2s) | 5 sigma gate (not a CI, see checkpoint) | 6000 shots x 2 quadratures, single seed pair (sub_seed cp5-re/cp5-im) | notebook cell 41 (implementation) + cell 43 (checkpoint), executed headlessly via nbconvert prefix run | 2026-08-12 R008 | Overnight IBM hardware job (tonight's gate, per c1.md) | job_id = `d9u95vs98n5s7392iao0`, backend = `ibm_marrakesh` (instance "Martin", open/free plan). Job status = DONE (checked 2026-08-13 before the 13:00 go/no-go). Analysis (R008b): hardware chi(0.9) = -0.1190 - 0.0200j vs exact +0.6623 - 0.1320j -- deviation 35.2 sigma (re), 5.0 sigma (im); signal survival 0.179. Consistent with decoherence at this circuit depth (1729, 435 CZ, Trotter reps=1) dominating the signal. The chi figures are ancilla-only and therefore basis-independent, so they stand. **CORRECTION (2026-08-13, see BUGLOG B04): an earlier version of this row also reported "‘_rho^(I) = +1.5945+-0.0402 vs exact +2.2675, 16.7 sigma" and attributed the gap to hardware noise. That line was WRONG and has been withdrawn.’** This job used ONE fixed shadow basis [X,Y,Z]; the shadow estimator presumes bases drawn i.i.d. uniform per shot, so with a fixed basis only Q's Z2 term can match and the estimator returns 3*, not ‘. Verified on the NOISELESS simulator: fixed-basis [X,Y,Z] gives +2.6970 vs 3’exact = +2.6964 (agreement to 4 dp) and 65.4 sigma from ‘, with no noise present at all. No shadow-derived SYSTEM observable can legitimately be reported from this job -- only chi | 5-sigma is the project's own acceptance bar (CONVENTIONS §6); 35 sigma is far outside it | 2 circuits x 2000 shots = 4000 shots | t=0.9, phi in {0, -pi/2}, basis=[X,Y,Z] on (q0,q1,q2), method="trotter" reps=1 (hardware-realistic per CONVENTIONS §2 U(t) law), transpiled depth 1729 / 435 CZ each; submitted via qiskit-ibm-runtime SamplerV2, live from the validated notebook kernel state (build_shadow_hadamard_circuit, HAM, sub_seed reused unmodified) | 2026-08-12 (submitted) / 2026-08-13 (analysed) R009 | Full time sweep, Challenge 5 `run_time_sweep` (given-complete,* exercised at official budget) | 3,456 circuits, 256,000 kept shots, 128 ShadowRecords retained, wall time 138.8s (run_summary.json) | n/a (raw sweep, validated in R010) | N_TIMES=64 x 2 quadratures x SHOTS=2000, seed=2026 (SEED), method="exact" | notebook cell 48, executed headlessly via nbconvert prefix run (.venv, kernel qh26-t5) | 2026-08-12 R010 | Checkpoint 6 (Challenge 5+6, given-complete, full official budget) | 8/8 PASS: 6.1 chi max dev 2.23 sigma (<5); 6.2 chi rms/sem=1.05 (in [0.5,1.6]); 6.3 =+0.0714+-0.0082 vs exact +0.0739 (0.3s); 6.4 ‘=+2.2708+-0.0052 vs exact +2.2675 (0.6s); 6.5 (t=4.2)=+0.4170+-0.0271 vs rho^(l) ref +0.4369 (0.7s); 6.6 chi_Q max dev 2.35 sigma; 6.7 scaling slope -0.463 (theory -0.5, gate +-0.15); 6.8 3 joint obs + /‘/ from 128 shared record sets; per-time ‘ flat within 5 sigma at every time point (max 2.2 sigma over 64 times, notebook cell 53). chi(t) rms error 0.0290 (CONVENTIONS calibration ~0.026); scaling study N=128/512/2048 x 24 repeats: mean|err|=0.0880/0.0392/0.0244 | 5 sigma gates (6.1,6.3-6.6); N^-1/2 gate +-0.15 (6.7); rms/sem ratio gate [0.5,1.6] (6.2) | 256,000 shots (main sweep) + 3x2x24x(~500 avg)=~72,000 shots (independent scaling study) | notebook cells 48,51,53,55,59, executed headlessly via nbconvert prefix run | 2026-08-12 R011 | Challenge 7 DFT baseline (`dft_peak_labels`), data-only | 3/4 lines found (E=-1.890,+0.539,+1.816 vs exact -1.915,+0.550,+1.946; E=-0.481 line missed entirely, as CONVENTIONS §5 predicts from the Hann half-lobe vs populated-spacing margin); q_ratio at the weak found line (E=+1.816) = +1.153 vs exact +1 | n/a (baseline, no CI) | reuses the 256,000-shot sweep (R009), no new shots | notebook cell 65, executed headlessly via nbconvert prefix run | 2026-08-13 R012 | Challenge 8 matrix-pencil reconstruction (`matrix_pencil`, `reconstruct`), data-only | all 4 populated lines recovered: E=-1.9180(+1),-0.4849(+1),+0.5516(+3),+1.9787(+1) vs exact -1.9152,-0.4810,+0.5500,+1.9463; p=0.1660/0.1518/0.6298/0.0506 vs exact 0.1597/0.1527/0.6337/0.0538; q_hat=+0.955/+0.974/+3.007/+0.855, all round to correct integer sector; rank auto-selected = 4 | not yet bootstrapped (see R013) | reuses R009's shots, no new shots | notebook cell 70, executed headlessly via nbconvert prefix run | 2026-08-13 R013 | Checkpoint 7 (Track A acceptance, Challenges 7-9 complete) | 8/8 PASS: 7.1 4 modes (>=3); 7.2 max |dE|=0.0325 (<0.05); 7.2b worst DFT-pencil distance 0.163 (<0.3, data-only cross-check); 7.3 max |dp|=0.0063 (<0.02); 7.4 all labels round correctly (+0.96,+0.97,+3.01,+0.85 -> +1,+1,+3,+1); 7.5 uncertainties finite/positive, weakest-line q_sd=0.187 > 2x strongest q_sd=0.016 (correctly uncertain where it should be); 7.6 pencil weak-line label +0.855 beats DFT ratio +1.153 (exact +1); 7.7 tripwire: reconstruct() reproduces identical output with 21 exact-reference names replaced by raising stubs (zero fabrication confirmed). Bootstrap: 200 replicates, parametric (perturb by sem, full pipeline rerun), E_sd/p_sd/q_sd = [0.0053,0.0053,0.0014,0.0167]/[0.0023,0.0025,0.0027,0.0031]/[0.052,0.063,0.016,0.187] | per-item gates: 7.2<0.05, 7.2b<0.3, 7.3<0.02, 7.5/7.6 structural, 7.7 exact match | reuses R009's 256,000 shots; bootstrap is pure post-processing (0 new shots, 200 replicates in 0.1s) | notebook cells 65,70,73,76,80, executed headlessly via nbconvert prefix run | 2026-08-13 R023 | **Going-further robustness item 4 — reduced 2-site instance on real hardware, with a GENUINE shadow ensemble — RETURNED, and it is the cleanest hardware result of the whole project** | Side model (NOT the frozen benchmark; robustness claims only): H2 = 0.65(XX+YY) + 0.25 ZZ + 0.40 Z0 - 0.50 Z1 on 2 system qubits + ancilla = 3 qubits; Q2 = Z0+Z1; prep Ry(1.3) on q0; spectrum {-1.8311, 0.15, 0.35, 1.3311}, [H2,Q2]=0 exactly. Transpiled on ibm_kingston: **depth 64, 15 two-qubit gates** — against the frozen benchmark's 101 (exact path) and 435 (Trotter, R008). At n=2 the FULL balanced basis ensemble is only 3^2=9 bases, so this is the first hardware job here that can report a **valid system observable** rather than chi alone (closes the BUGLOG B04 restriction). Job `d9uk99k98n5s7392vhsg`, ibm_kingston, 18 circuits x 2000 shots = 36,000 shots. PREDICTED before submission, hence falsifiable: chi survival (1-1.99e-3)^15 = **0.971**; should land within a few sigma of exact +1.2675. Ideal-simulator validation of the identical construction (72,000 shots): chi 0.8/1.8 sigma, = +1.2706+-0.0083 vs exact +1.2675 (0.4 sigma) — so the estimator chain is verified before any QPU time was spent. **RESULT (2026-08-13): chi(0.9) = +0.6078+0.1364j vs exact +0.6345+0.1277j -- 4.5/1.2 sigma, chi survival 0.962. from GENUINE random-basis shadows = +1.2127+-0.0119 vs exact +1.2675 -- 4.6 sigma, survival 0.957.** Both well within the project's usual 5-sigma acceptance bar, on ACTUAL noisy hardware, not simulator. This is the single result that most directly demonstrates the project's core claim end to end: a properly randomized shadow-Hadamard protocol recovers a valid system observable from real QPU data, closing BUGLOG B04 with a positive result rather than only the negative one (fixed-basis jobs cannot do this) | measured 4.5/1.2/4.6 sigma, all <5 | 36,000 shots (~12 s charged QPU) | `python hardware_run.py --submit --backend ibm_kingston --model 2site --shadow`;

*validated first by evidence/scripts/validate_2site.py; fetched via `hardware_run.py --fetch d9uk99k98n5s7392vhsg` | 2026-08-13 R024 | **`packing.py` is NOT USABLE** — third-party file, wrong dependency, never part of this repo | `packing.py` (copyright Jiun-Cheng Jiang, Apache-2.0) appeared untracked in the project root; `git log --all -- packing.py` shows zero commits ever touched it. It imports `.layout` (`DeviceProfile`, `tile_disjoint`) from a sibling module that does not exist anywhere on this machine — confirmed by direct import, `ModuleNotFoundError/ImportError`. Not used, not completed, not shipped. Its GOAL (pick the best-connected, lowest-error physical qubit window) was instead solved with stock Qiskit + rustworkx in `layout_search.py`, self-contained, no foreign dependency. **CORRECTION (2026-08-13, B08): the "never part of this repo" claim briefly went false** — a later blanket `git add` accidentally tracked the file; caught by pre-push audit, `git rm --cached` applied, and it is gitignored going forward, restoring the state this row describes | n/a | n/a | `python -c "import packing"` reproduces the import failure | 2026-08-13 R025 | Local transpile cost check: does the exact-path advantage survive to n=4? | **CORRECTED by C4 audit (2026-08-13) — original row was unscripted, not reproducible, and its n=3/opt=1/exact figure (92) diverged from R018's independently-verified 101 for the nominally same circuit.** Root cause found and fixed: the original check used an arbitrary seed_transpiler (1234), not `sub_seed("hw-transpile")` (the actual value R018's real submission used, =2020549847 under SEED=2026). Re-run with the correct seed, now scripted and reproducible (evidence/scripts/transpile_cost_n3_n4.py), against REAL kingston coupling map: n=3/opt=1/exact = **101 two-qubit gates, MATCHES R018 exactly**. Full corrected table (kingston, seed_transpiler=sub_seed("hw-transpile")): n=3 opt=1 exact=101/trotter=435; n=3 opt=3 exact=70/trotter=239 (**3.41x** exact advantage); n=4 opt=1 exact=476/trotter=668; n=4 opt=3 exact=324/trotter=376 (**1.16x**, advantage nearly gone) — same qualitative conclusion as before (consistent with CONVENTIONS §2's O(4^n) crossover warning), numbers now corrected and self-consistent with R018. SEPARATE FINDING, unaffected by the correction: optimization_level=3 vs 1 alone gives 101->70 (n=3 exact) and 435->239 (n=3 trotter) with no other change — R008/R018 used opt=1; hardware_run.py default changed to opt=3 as a result | n/a (deterministic transpiler comparison) | 0 (local transpile only, no QPU) | evidence/scripts/transpile_cost_n3_n4.py — run `python evidence/scripts/transpile_cost_n3_n4.py`; its own final line cross-checks against R018's confirmed 101 | 2026-08-13 R026 | **4-site side model, best-layout hardware test — RETURNED, and it disagrees sharply with an independent run on the IDENTICAL layout** | Extended side model (NOT the benchmark, robustness claims only, NOT checkpoint-validated like 2site — see R024's replacement tool note): 4 system qubits + ancilla = 5 total. `layout_search.py` evaluated 10 candidate physical-qubit windows on ibm_kingston, greedily seeded from the lowest-calibrated-error edges (352 directed coupling edges, median 2q err 2.08e-3); scored each by the ACTUAL transpiled circuit's real gate list x each used edge's real calibrated error (not a device-wide median). Best: window (124,125,126,127,137), 386 two-qubit gates, **predicted survival 0.6511** (per-edge product; the CLI's coarser median-based estimate said 0.464 for the same job — both recorded, falsifiable). Job `d9ultm0u5hac73ahd9kg`, 2 circuits (fixed basis, chi only per B04), 4000 shots. Ideal-simulator validation BEFORE submission (evidence/scripts/validate_4site.py, 486,000 shots): [H,Q]=0 exactly, chi 0.2/1.3 sigma, ‘ 0.4 sigma from exact — PASS. **RESULT (2026-08-13): chi(0.9) = +0.1790-0.0480j vs exact +0.5560-0.3800j -- 17.1/14.9 sigma, survival 0.275.** Both predictions (0.6511 per-edge, 0.464 median) were optimistic; measured is worse than either. **STRIKING CROSS-CHECK: a teammate independently submitted the EXACT SAME model+layout+backend (R031, job d9umbk498n5s739327k0) and got survival 0.587 -- more than 2x ours, on a nominally identical circuit.** Job creation timestamps checked: ours 2026-08-13T06:06:17Z, theirs 2026-08-13T06:36:00Z (UTC) -- only ~30 minutes apart, ruling out slow multi-hour/day calibration drift as the explanation. This is genuine SHORT-TIMESCALE run-to-run hardware variance: the same device, same circuit, same layout, half an hour apart, gave results differing by more than a factor of 2. Reported as-is -- neither number is "the" 4-site result, both are single data points from a noisy process, and that variance is itself the finding | two survival predictions recorded pre-return: 0.6511 (per-edge) and 0.464 (median); measured 0.275 vs independent replicate 0.587 | 4,000 shots hardware (~3 s QPU); 486,000 shots ideal-sim validation | `python layout_search.py --backend ibm_kingston --model 4site --candidates 10` then `python hardware_run.py --submit --backend ibm_kingston --model 4site --initial-layout "124,125,126,127,137"`; fetched via `hardware_run.py --fetch d9ultm0u5hac73ahd9kg` | 2026-08-13 R021 | **Ablation A — protocol x estimator grid, with the standard-vs-shadow equivalence measured** | (a) EQUIVALENCE CONFIRMED: a genuine textbook Hadamard test (ancilla only, system columns discarded) run over the full 64-point grid* *at identical budget gives chi rms error 0.0265 vs shadow 0.0290, and* ***identical mean sem 0.0277 vs 0.0277****; |chi_std - chi_shadow|/combined sem = max 2.10, mean 0.88 -> statistically the SAME random variable, as Checkpoints 3b/3c predict. Measuring the "garbage" register costs the ancilla signal nothing. (b) 2x2 GRID: standard+DFT 3/4 lines, max|dE| 0.1387, labels IMPOSSIBLE; standard+pencil 4/4, max|dE| 0.0126, labels IMPOSSIBLE; shadow+DFT 3/4, max|dE| 0.1307, labels available; shadow+pencil 4/4, max|dE| 0.0325,* ***labels 4/4 correct****. The axes are orthogonal: ESTIMATOR choice buys line recovery (3/4 -> 4/4, ~10x better dE), PROTOCOL choice buys the symmetry labels, and no post-processing can extract q from chi alone. (c) COST: conventional route needs* ***13 dedicated modified Hadamard tests*** *(Q=3, H=9, Z0=1 Pauli terms) for the same three joint observables; shadow route needs* ***0 extra circuits****, paying instead a measured variance premium of only* ***1.07x*** *on chi_Q (sem 0.1012 measured vs 0.0949 predicted for 3 dedicated tests sharing the same shots) — while also delivering H and Z0, which would have cost 10 further experiments | equivalence z-scores as stated; cost factor from measured vs analytic sem | standard-Hadamard arm 64 x 2 x 2000 = 256,000 new shots; shadow arm reuses R009 | evidence/scripts/ablation_a_b.py driving graded code via nbclient | 2026-08-13 R022 |* ***Ablation B — resolution failure boundary (Going-further Track A item 4, T_max half)*** *| Truncating the same series to shorter T_max: the* ***DFT never resolves the 1.031-spaced pair (E=-0.481 vs +0.550) at ANY T_max up to the full 12.6****, exactly as CONVENTIONS §5 predicts from the Hann half-lobe (0.997) vs spacing (1.031). The* ***matrix pencil resolves it from T_max >= 7.0*** *and recovers all 4 lines with all 4 labels correct from T_max >= 7.8. So the pencil achieves with* ***56% of the time grid*** *what the windowed DFT fails to achieve with 100% of it. DFT line-count is also non-monotonic in T_max (1,1,1,1,1,2,2,2,1,1,2,2,3,3,3 over M=8..64), i.e. its peak-finding is unstable, not merely blunt | boundary resolved to the 4-point grid step (dt=0.2, so +-0.8 in T_max) | pure post-processing of R009's 256,000 shots; 0 new shots | evidence/scripts/ablation_a_b.py, 15 truncation points M=8..64 | 2026-08-13 R019 |* ***12-seed Part-B reproducibility study — closes the >=10-seed house-law gap*** *| 12/12 independent full-budget sweeps recovered rank 4 and all 4 populated lines;* ***12/12 seeds got every symmetry label correct after rounding****. Across-seed mean +- sd (95% CI = 1.96 sd per CONVENTIONS §7): E = -1.9146+-0.0064 (CI +-0.0125), -0.4849+-0.0051 (+-0.0100), +0.5501+-0.0020 (+-0.0038), +1.9481+-0.0127 (+-0.0248) vs exact -1.9152/-0.4810/+0.5500/+1.9463; p = 0.1599+-0.0022, 0.1528+-0.0029, 0.6323+-0.0037, 0.0528+-0.0027 vs exact 0.1597/0.1527/0.6337/0.0538; q = +1.002+-0.068, +1.013+-0.043, +3.009+-0.016, +0.967+-0.179. Estimator is unbiased to within ~1 sd on every quantity (largest bias: E of the -0.481 line, -0.0039 ~ 0.8 sd) | across-seed sd over 12 independent seeds; 95% CI = 1.96 sd as tabulated | 12 x 256,000 = 3,072,000 shots, declared SEPARATELY from the 256,000-shot headline budget (same pattern the notebook uses for its 24-seed scaling study) | evidence/scripts/seed_sweep.py driving the graded run_time_sweep + reconstruct via nbclient; CSVs written to data/ per CONVENTIONS §7 | 2026-08-13 R020 | Bootstrap validation: is the R013 parametric bootstrap honest? | Ratio of (across-seed sd, R019) to (bootstrap sd, R013) per line —* ***E: 1.20, 0.97, 1.35, 0.76 ; q: 1.32, 0.68, 1.03, 0.96****. All ratios lie in 0.68-1.35, i.e. the parametric bootstrap is a faithful proxy for real seed-to-seed spread; worst case it UNDERSTATES by ~35% (E of the +0.55 line, ratio 1.354; and q of the -1.92 line, 1.32). With only 12 seeds the across-seed sd carries ~21% relative uncertainty of its own (1/sqrt(2(n-1))), so every ratio is consistent with 1.0 within the comparison's own noise. CONCLUSION: the R013 error bars are validated, not merely assumed — no evidence they are materially wrong, and no widening required before use | ratios as tabulated; comparison noise ~21% | reuses R013 (256,000 shots) and R019 (3,072,000 shots); pure post-processing | evidence/scripts/seed_sweep.py final block | 2026-08-13 R018 |* ***Hardware 2x2 ablation {trotter, exact} x {marrakesh, kingston} — ALL 4 ARMS COMPLETE*** *| Completes the grid around R008 to separate DEVICE quality from CIRCUIT DEPTH empirically. Transpiled cost (optimization_level=1, identical seed): exact path depth 375 /* ***101*** *two-qubit gates; trotter reps=1 depth 1729 /* ***435*** *two-qubit gates -> exact is* ***4.3x shallower on real hardware****. Jobs: exact+kingston* `d9ujlb0u5hac73ahadu0` *(16 queued); exact+marrakesh* `d9ujlc0u5hac73ahadvg` *(22 queued); trotter+kingston* `d9ujld498n5s7392uqeg` *(17 queued); reference arm trotter+marrakesh = R008* `d9u95vs98n5s7392iao0` *(measured damping 0.179). PREDICTED survival from measured 2q error rates (kingston 1.99e-3, marrakesh 3.23e-3): trotter 0.420 / 0.245, exact 0.818 / 0.721 — recorded BEFORE results return so the prediction is falsifiable.* ***TWO OF THREE ARMS RETURNED (2026-08-13):*** *exact+marrakesh: chi=+0.5550-0.0210j, survival* ***0.822****, 5.8/5.0 sigma. exact+kingston: chi=+0.5820-0.1120j, survival* ***0.878****, 4.4/0.9 sigma — the BEST result of any hardware job this session, on the device independently confirmed best-calibrated by layout_search.py (see R026). Committed evidence for both fetches:*

evidence/hw_2x2_fetch_log.txt (C4 audit: this row previously had no durable evidence, only prose — fixed). Against the same-device Trotter reference (R008, survival 0.179, 35.2 sigma) this is a SINGLE-RUN comparison CONSISTENT WITH depth being the dominant factor: signal survival improved 4.6x (435 -> 101 two-qubit gates) on marrakesh, the one device where both circuit variants have now actually been measured. Kingston's own Trotter arm is still queued, so the depth effect is demonstrated on ONE device so far, not yet reproduced across both (C4 audit: earlier text incorrectly said "reproduced on BOTH devices" while the same sentence noted kingston's Trotter arm was still pending — corrected). Device axis, exact path held fixed: marrakesh 0.822 vs kingston 0.878, a further 1.07x consistent with picking the better-calibrated device. Naive (1-p)^n_2q was CONSERVATIVE on both arms — predicted 0.721/0.818, measured 0.822/0.878, independent-per-gate depolarising underestimates real performance here by ***marrakesh (0.822-0.721)/0.721=14.0%, kingston (0.878-0.818)/0.818=7.3%*** *(C4 audit: previously stated as "~10-14%"; corrected range is ~7-14%).* ***FOURTH ARM RETURNED (2026-08-13, job `d9ujld498n5s7392uqeg`): trotter+kingston chi=-0.1820+0.1690j, survival 0.368, 38.4/13.7 sigma.*** *THE GRID IS NOW COMPLETE: || marrakesh | kingston ||---|---|---|| trotter | 0.179 | 0.368 || exact | 0.822 | 0.878 |. Depth effect (exact/trotter) at fixed device: marrakesh 0.822/0.179=**4.59x**, kingston 0.878/0.368=**2.39x** -- bigger on the noisier device. Device effect (kingston/marrakesh) at fixed circuit: trotter 0.368/0.179=**2.06x**, exact 0.878/0.822=**1.07x** -- bigger on the deeper circuit. Neither is a contradiction: both are the expected qualitative signature of a multiplicative per-gate error model (survival ~ (1-p)^n_gates) -- removing gates matters more when p is larger, and device quality matters more when n is larger. Depth remains the dominant lever throughout (2.4-4.6x) vs device (1.1-2.1x), confirming the original claim now on a complete grid rather than 3/4 of one | measured vs predicted as stated | 4 jobs x 2 circuits x 2000 shots = 16,000 shots | evidence/scripts/submit_hw_2x2.py driving the graded build_shadow_hadamard_circuit via nbclient (cells 0-22 prefix); t=0.9, phi in {0,-pi/2}, basis [X,Y,Z] — every parameter except method/backend held identical to R008; raw fetch output in evidence/hw_2x2_fetch_log.txt | 2026-08-13 R017 |* ***EXPLORATORY — CONFOUNDED, NOT SLIDE-READY*** *— "Going further (robustness)" item 2: does the symmetry label survive noise? | (a) WEIGHT-DAMPING CLAIM CONFIRMED: p_noisy/p_ideal = 0.352 +- 0.026 across matched modes, vs the independently measured mean|chi| damping 0.342 -- weights are biased low by exactly the damping factor. (b) LABEL-SURVIVAL CLAIM NOT ESTABLISHED: q shifts by dq = -0.072 (E=-0.48 line) and -0.449 (E=+0.55 line); labels correct after rounding ideal 2/2 vs noisy 2/3; the Q=3 line landed 0.449 from its ideal value, i.e. within 0.05 of flipping to the wrong integer. Partial cancellation is visible (p fell to 35% while q moved 2-14%) but "largely cancels" is too strong at this budget. (c) SEPARATE FINDING: noise inflates the selected rank 3 -> 7, manufacturing 4 spurious modes (E=-4.39,-3.61,+1.20,+2.45 correspond to no exact line). CONFOUND (why this is not reportable): the ideal-Trotter baseline is itself broken on this reduced grid -- it recovers only 3 modes, MISSES two of the four exact lines (-1.9152, +1.9463), places the -0.4810 line at -0.7582 (off by 0.28) and invents a spurious -3.5202. Reduced grid (22 pts, dt=0.60, 800 shots) + trotter reps=1 (operator error 0.822 per R003) degrade the reconstruction before noise is added at all, so label degradation cannot be cleanly attributed to noise. A clean test needs the full grid/exact path, i.e. re-running the noise sweep at Part-B fidelity | n/a — exploratory, single seed, confounded baseline; explicitly NOT a supported claim | reuses the Challenge-11 noise sweep (22 times x 2 quadratures x 800 shots x 2 backends); zero new circuits beyond it | evidence/scripts/robustness_item2.py appended to the cells 0-82+89-93 prefix, executed headlessly (scratch only — the graded notebook was NOT modified) | 2026-08-13 R016 | Challenge 10 bonus (`krylov_lowest_energy`, given-complete driver, post go/no-go per user "go ahead") | E0_krylov = -1.5349 (m=16, threshold=0.02, 4 vectors kept) vs exact lowest POPULATED level -1.9152 (error 0.3803); vs unpopulated global ground -2.6350 (correctly not converging there); matrix-pencil comparison (R012/R013) -1.9180, much closer -- expected, Krylov is the harder/ill-conditioned method here. Advisory checkpoint PASS: |E0-populated|=0.38<0.6 and |E0-global_ground|=1.10>0.3. Threshold/m regularisation sweep (26 thresholds x 4 dims = 104 GEVP solves) ran without error, reproducing the notebook's own documented instability shape | soft/advisory only (CONVENTIONS §6 STRICT_CHECKS: soft checks never fail the notebook; this GEVP is genuinely stochastic run to run, explicitly not held to a hard bound) | reuses R009's 256,000 shots (chi, chi_H already recorded -- zero new circuits, the whole point of the bonus) | notebook cells 85,87 executed headlessly via nbconvert prefix run | 2026-08-13 R014 | Challenge 11 noise robustness (given-complete, default params) vs. real hardware (R008) | simulated noisy/ideal mean|chi| damping ratio = 0.342 (p1=3e-4, p2=6e-3, p_ro=1.2e-2, method=trotter reps=1, matches "the noise model actually does* *something" soft-check PASS); real ibm_marrakesh damping ratio = |chi_hw(0.9)|/|chi_exact(0.9)| = 0.1206/0.6754 = 0.179 (from R008). Real hardware attenuates ~2x more than the notebook's generic depolarizing+readout toy model predicts at this circuit depth (1729/435 CZ, Trotter reps=1) | n/a (single-point ratio comparison, not a CI claim) | simulated: N_NOISE~22 x 2 quadratures x 800 shots (ideal + noisy sweeps); real: reuses R008's 4000 shots | notebook cells 91,93 (given-complete, unmodified) executed headlessly via nbconvert prefix run, skipping cells 83-88 (Challenge 10 Krylov, correctly untouched -- post-freeze-only per CLAUDE.md) | 2026-08-13 R041 | **Three-arm observable reconstruction: which protocol can rebuild which observable, each with its proper analysis chain** | Arms: (1) GARBAGE DISCARDED -- Hadamard test only, chi -> windowed DFT; (2) POST-SELECT all-Z -- same shots, Z-only subset, matrix pencil; (3) CLASSICAL SHADOW -- full ensemble, matrix pencil. Target observables Z_0, Z_0Z_1, H, and the hopping term X_0X_1+Y_0Y_1. Per-line labels compared against the quantity the estimator actually targets, / (NOT ; the two coincide only when [O,H]=0 -- getting that wrong was a reference bug on the first run, corrected before reporting). **ARM 1 (discard):** 3/4 energies via DFT at E=-1.890/+0.539/+1.816, and **ZERO observables** -- the system register was never measured, so no O can be formed at any shot count. Structural, not precision. **ARM 3 (shadow):** reconstructs ALL FOUR. Z_0 labels [-1.013,-1.004,+1.009,-1.048] vs exact [-1,-1,+1,-1], **max err 0.048**; Z_0Z_1 [-1.057,-0.993,+0.994,-0.951], **max err 0.057**; H [-1.793,-0.416,+0.539,+2.085] vs [-1.915,-0.481,+0.550,+1.946], **max err 0.139**; hopping [-1.611,+0.542,-0.026,+4.658] vs [-1.793,+0.414,0,+4.148], **max err 0.509** (largest values, so relatively comparable). **ARM 2 (post-select):** Z_0 and Z_0Z_1 are 1/1 Z-diagonal so it CAN reconstruct them -- but at only 3.7% shot yield the weakest line degrades badly (recovered -0.594 vs exact -1.000, ~40% error). H is 5/9 diagonal -> **SYSTEMATICALLY BIASED, max 2.696** (recovered -0.819 where -1.915 was wanted); more shots do not help, since it converges to Tr[O_Zdiag U rho], a different quantity. The hopping term is 0/2 diagonal -> **IMPOSSIBLE**, no unbiased estimator exists. **HONEST SCOPE OF THE CLAIM:** "only the classical shadow can reconstruct these" is TRUE for H and X_0X_1+Y_0Y_1 against BOTH other arms, and TRUE for Z_0/Z_0Z_1 against the discard arm -- but FALSE for Z_0/Z_0Z_1 against post-selection, which handles Z-diagonal observables correctly (and per R039 a dedicated Z sweep does so more cheaply than shadows). The exclusive capability is non-commuting observables, not diagonal ones | label errors as tabulated; post-select bias is systematic, quantified | 256,000 shots (one shadow sweep re-analysed three ways) | evidence/scripts/observable_reconstruction.py; evidence/observable_reconstruction.json; evidence/observable_reconstruction_console.txt | 2026-08-13 R040 | **The protocol escalation ladder — four ways to treat the garbage register, and what the shadow inversion actually contributes** | Extends R039 with the two rungs it was missing: the discard-the-garbage baseline (the plain Hadamard test) and, more interestingly, **post-selecting the accidentally-all-Z shots out of the SAME shadow experiment** -- the check that tests whether the 3^w inversion is doing real work or is just randomisation plus post-selection. All four measured on one grid, 64 times x 2 quadratures x 2000 shots. **PRECISION on chi_Q:** (0) discard garbage -- **IMPOSSIBLE**, no system data at all; (1) post-select all-Z from the shadow run -- sem **0.3400**, keeping only **3.7%** of shots (matches the predicted 1/3^3 = 3.7% exactly); (2) dedicated all-Z sweep -- sem **0.0650** (R039); (3) shadow inversion -- sem **0.1012**. **THE KEY RESULT: the shadow inversion beats naive post-selection of its own data by 3.36x** (0.3400 -> 0.1012), and the reason is structural -- each Pauli TERM Z_j of Q only requires b_j = Z, i.e. 1/3 of shots, whereas post-selection demands all n qubits be Z simultaneously, i.e. 1/3^n. The inversion recovers exactly the shots post-selection throws away. rms vs exact: shadow 0.1008, post-select 0.3225. **CAPABILITY (what each can measure at all, independent of precision):** discard -> chi only; both Z-only rungs -> chi and Q but only **5 of 9** terms of H, hence the conserved-quantity drift of 0.147 documented in R039 that they cannot self-detect; shadow -> all nine terms, every observable, no advance commitment. **HONEST ORDERING:** a dedicated Z experiment still beats the shadow by 1.56x on Q alone. The ladder's point is not that shadows win everywhere -- it is that each rung buys a capability the one below cannot have at any shot count | sems and yields as measured; capability rows are structural, not statistical | 256,000 shots (one shadow sweep, re-analysed four ways); dedicated-Z arm reuses R039 | evidence/scripts/protocol_escalation.py; evidence/protocol_escalation.json | 2026-08-13 R039 | **Direct Z-basis readout vs the shadow ensemble — the motivation ablation, and the answer does NOT flatter shadows** | Asks the sceptic's question: are classical shadows needed AT ALL for symmetry resolution? Q = sum_j Z_j is DIAGONAL in the computational basis, so it can be read straight off every shot with no randomisation and no 3^w inversion. Fixed all-Z readout, 1 circuit per (t,phi) instead of up to 27, same 64-point grid and 2000 shots/setting. **(1) FOR Q ALONE, DIRECT Z***

IS BETTER, and we report that: direct-Z chi_Q rms error 0.0669 with mean sem **0.0650**, against the shadow ensemble's mean sem **0.1012** (R021) -- so the shadow route pays a **1.56x variance premium on the symmetry channel it is supposed to be good at**. Reason is structural, not statistical noise: every shot contributes to a direct-Z estimate, whereas the shadow estimator's indicator 1[b_j = P_j] discards any shot whose drawn basis does not match; and the per-shot second moment is E[Q^2] <= 9 versus the shadow's <= 15. chi itself is ancilla-only and identical either way (rms 0.0294). **(2) WHAT DIRECT Z STRUCTURALLY CANNOT DO -- the actual motivation:** a Z-only experiment sees only the Z-diagonal part of any observable. For our H that is **5 of 9 Pauli terms**; the 4 invisible ones are exactly the hopping terms IXX/IYY/XXI/YYI, each with the LARGEST coefficient 0.65. The consequence is not a small bias: is CONSERVED and must be flat in t, but **drifts by up to 0.1466** over the grid (+0.0739 exact vs +0.2205 at t=2.1). So a Z-only experiment reports a time-varying wrong answer for a conserved quantity and, having no second channel, **cannot detect that it is wrong** -- whereas our shared-record design cross-checks the ancilla channel against the system channel (cf. R033). **CONCLUSION, stated as found:** the honest case for classical shadows is NOT that they resolve the symmetry better -- direct Z does that more cheaply. It is that the SAME shots also yield observables that do not commute with Q, with no advance commitment to which ones. Premium 1.56x on the symmetry channel; return is every other observable | sems as measured; drift values exact (evaluation side) | 64 times x 2 quadratures x 2000 = 256,000 shots, direct-Z arm; shadow comparison reuses R021 | evidence/scripts/direct_z_ablation.py; evidence/direct_z_ablation.json | 2026-08-13 R038 | **PROTOTYPE (next-phase, NOT a Topic-5 deliverable) — symmetry-resolved DENSITY OF STATES from the unmodified shadow-Hadamard pipeline** | Motivated by a second paper (arXiv 2407.03414v2, DOS for fermionic systems) proposed as a follow-on. **MAPPING NOW VERIFIED AGAINST THE PAPER (paper/2407.03414v2.pdf, read 2026-08-13).** It was derived independently first, then checked, and it matches: their Eq. (7) is G(t) = (1/sqrt(2pi)) Tr[e^{-iHt}] -- the FDOS is the trace of the propagator, exactly the substitution made here; their **Statement 2** shows a unitary 1-design suffices and proposes {1,X}^n, i.e. random bit-flips giving random COMPUTATIONAL BASIS states, of which our exhaustive 8-state enumeration is the deterministic version; and their Fig. 2 caption sets W=H for the real part and W=S^dag H for the imaginary -- identical to our PHI_RE/PHI_IM quadrature convention. So this row is a valid partial reimplementation of their core protocol, independently arrived at. The mapping: the DOS is the Fourier transform of Tr[U(t)] (the TRACE, not a state expectation), while our chi(t) = Tr[U(t) rho]; so setting rho = I/2^n converts the same experiment into a DOS estimator, and our chi_Q channel converts it into a CHARGE-resolved DOS -- for a fermionic system under Jordan-Wigner that is a particle-number-resolved DOS, since N = (n-Q)/2. **ONLY `prep` CHANGED** -- same circuit builder, parse_memory, three estimators, matrix pencil, all untouched. Mixed state realised by exact enumeration over all 2^n=8 computational basis states (Tr[U] = sum_z) rather than random-state typicality, whose O(2^(-n/2))=35% error is useless at n=3. Budget 8 preps x 32 times x 2 quadratures x 500 shots = 256,000 shots. **RESULT — the machinery transfers:** chi rms error 0.0164 vs the exact trace signal, chi_Q rms 0.0730; chi(0) = +1.0000 exactly (= Tr[I]/2^n) and chi_Q(0) = +0.0450 vs exact 0.0000 (= Tr[Q]/2^n, Q traceless). Unmodified `reconstruct()` returned rank 6 from the 8-level full spectrum. **THE DECISIVE SIGNATURE — peak weights become DEGENERACIES/2^n, not state populations:** the five isolated levels gave weights 0.1256/0.1272/0.1201/0.1119/0.1090 against the predicted 1/8 = 0.125, and the three-fold merged cluster near E~0.49 gave **0.3541 against the predicted 3/8 = 0.375**; total weight 0.9478 vs the expected 1.0. Isolated energies accurate to 0.012 (worst: -2.6468 vs exact -2.6350). **PREDICTED FAILURE OCCURRED EXACTLY AS PREDICTED, and was recorded before the run:** the triple {0.4500, 0.4636, 0.5500} (gaps 0.0136 and 0.0864, far below the Rayleigh limit 2pi/T_max = 0.5067) merged into one peak. **UNPREDICTED SUCCESS:** the pair {1.6215, 1.9463}, gap 0.3248, also sits BELOW the Rayleigh limit yet was resolved into two separate peaks -- matrix-pencil super-resolution, consistent with R022's independent finding that the pencil beats the windowed-DFT resolution bound. **WHERE WE CAN IMPROVE ON THE PAPER (the actual contribution, not the reimplementation).** Their headline fermionic innovation is the SUBSPACE DOS g_M(E) at fixed particle number, obtained (Statement 3, Eq. 13) by sampling initial states INSIDE each subspace -- Dicke states or fixed-Hamming-weight basis states -- which costs **one experiment per particle-number sector M** (their Fig. 1 plots M=1..11 separately). Our shadow channel gets every sector from ONE full-Hilbert-space run: chi_Q labels each DOS peak by charge, and because shadows yield ANY Pauli observable from the same shots we can also form chi’’’’’{Q^2}, chi_{Q^3}. FEASIBILITY CHECKED, not asserted: Q^2 and

Q^3 are only **4 Pauli terms each, max weight 3**, all measurable from the existing records, and the charge-moment Vandermonde over Q in {-3,-1,+1,+3} has **condition number 31** -- comfortably invertible. So the moments at each peak determine the full distribution of particle-number sectors contributing to it, recovering all g_M(E) simultaneously where the paper needs n+1 separate experiments. That is this project's 'many quantities, one record set' thesis applied to their problem. NOT YET IMPLEMENTED -- the Q^k sweep and moment inversion are next-phase. STATUS: prototype demonstrating the pipeline transfers; NOT part of the Topic-5 deliverable | weights vs degeneracy/2^n as tabulated; no CI (single seed, prototype) | 256,000 shots, simulator only | evidence/scripts/dos_prototype.py; evidence/dos_prototype_result.json; evidence/dos_console_output.txt; cached signals evidence/dos_signals.npz | 2026-08-13 R037 | **SKQD-style sample-based diagonalisation on real hardware shots, with symmetry-based configuration recovery — SCOPE-LIMITED BY DESIGN | What is NOT claimed, stated first:** SKQD's value proposition is spanning a subspace too large to enumerate. Our reachable space is 3-dimensional (2site) / 4-dimensional (frozen), so sampling finds every configuration, the subspace is COMPLETE, and exactness is guaranteed by dimension counting rather than by the method working. That advantage is out of reach at this benchmark size and is **not** demonstrated. What IS testable and is the point: (a) the DATA PATH — the all-Z subset of our random-basis shadow ensemble genuinely is a set of computational-basis configurations sampled from rho^(I)(t), so our records feed an SKQD pipeline unmodified; (b) CONFIGURATION RECOVERY on genuinely noisy hardware data. **IDEAL-SIM CONTROL (recovery must change nothing): 2site 0/4,000 violating, raw and recovered both dim 3, eigenvalues [-1.8311, 0.15, 1.3311] = exact populated levels, 0 spurious both ways; frozen 0/4,000 violating, both dim 4, [-1.9152,-0.481,+0.55,+1.9463] = exact, worst |dE| 2.2e-16, 0 spurious both ways. REAL HARDWARE (ibm_kingston 2site, job d9uk99k98n5s7392vhsg = R023): 66/4,000 shots (1.650%, matching R036) violated charge conservation. RAW subspace dim 4 (configs [0,1,2,3]) -> eigenvalues [-1.8311, 0.15,** 0.35**, 1.3311], i.e. ONE SPURIOUS eigenvalue at +0.35 leaking in from the unpopulated q=-2 sector purely because noise corrupted 66 shots into a forbidden configuration. RECOVERED subspace dim 3 (configs [0,1,2]) -> [-1.8311, 0.15, 1.3311], exactly the populated levels, SPURIOUS COUNT 1 -> 0.** So on real data, symmetry-based configuration recovery removed exactly the artefact hardware noise introduced, with no reference state and no simulation — the same reference-free principle as R036, now shown to affect a downstream spectral estimate rather than only an error rate | eigenvalues exact to 0 / 2.2e-16 in all legs (the subspace is complete, so this is expected and is not evidence of method quality) | reuses R023's 36,000 hardware shots and 36,000/108,000 ideal-sim shots; 0 new QPU time | evidence/scripts/skqd_pipeline.py; evidence/skqd_result.json; evidence/skqd_console_output.txt | 2026-08-13 R036 | **Going-further robustness item 3 — sector post-selection as REFERENCE-FREE error detection, measured on real hardware** | Q = sum_j Z_j commutes with H, and the input state populates only a subset of charge sectors, so any all-Z-basis shot landing in a FORBIDDEN sector is a **detected error certified by symmetry alone** -- no reference state, no simulation, no exact diagonalisation, and the argument survives to any system size. This is SQD's "configuration recovery" specialised to our conserved charge; SKQD proper was deliberately NOT implemented (see note below). Sectors computed not assumed: 2site populated {0,+2} forbidden {-2}; frozen populated {+1,+3} forbidden {-3,-1}. **IDEAL SIMULATOR CONTROL (must fire at zero): 2site 0/4,000 violating = 0.000%; frozen 0/4,000 = 0.000%** -- detector correctly silent when there is nothing to detect. **REAL HARDWARE (ibm_kingston, 2site balanced shadow ensemble, job d9uk99k98n5s7392vhsg = R023): 66 of 4,000 all-Z shots violated charge conservation -> DETECTED ERROR RATE 1.650%.** Consistency cross-check: kingston's mean readout error 2.62e-2 gives P(>=1 of 2 system qubits flips) = 5.17%, and only flips that CHANGE the charge sector are detectable, so a detected rate comfortably below the raw flip probability is what one expects -- 1.65% vs 5.17% is consistent, not anomalous. **YIELD/ACCURACY TRADE-OFF, stated not hidden:** charge is only defined on all-Z shots, i.e. 1 setting in 3^n (11.1% of shots at n=2, 3.7% at n=3), so the diagnostic is free in circuits but its statistical precision is set by that reduced subset; and it is blind to any error that preserves charge (e.g. a phase error inside a sector). **DELIBERATELY NOT DONE:** feeding post-selected data back into the shadow estimators, which would BIAS them -- the estimators' unbiasedness rests on a uniform basis distribution, and discarding a symmetry-violating subset of only the all-Z shots breaks exactly that (same failure class as BUGLOG B04). Reported as a diagnostic only | ideal-sim control 0.000% on 4,000 all-Z shots each; hardware 66/4,000 | reuses R023's 36,000 hardware shots and 36,000/108,000 fresh ideal-sim shots; 0 new QPU time |

evidence/scripts/symmetry_error_detection.py; evidence/symmetry_detection_result.json; evidence/symmetry_detection_console.txt | 2026-08-13 R035 | **AQC-Tensor circuit compression, CERTIFIED BY OUR OWN TRACK-B A/B TESTER — and the answer is nuanced, not a clean win** | Approximate Quantum Compiling (qiskit-addon-aqc-tensor 0.3.1, quimb+jax MPS backend) applied to the time evolution at t=0.9, optimising a shallow ansatz (seeded from a reps=1 Trotter circuit) against an MPS target built from a deep reps=3 Trotter circuit. L-BFGS-B converged in <1 s both times. **THE NOVEL PART: the compression is certified by the Track-B observable (R034), not only by a classical fidelity** -- setting W = AQC circuit and U = exact evolution makes chi_AB = literally the compression fidelity, so it is MEASURABLE FROM SHOTS and would remain available on hardware at sizes where classical verification is impossible. The compressor and its certificate come from two different halves of this project. **UNCONTROLLED evolution (AQC is excellent here):** n=3 Trotter 68 CX / exact-synthesis 19 CX / **AQC 9 CX** at infidelity 1.4e-4; n=4 Trotter 104 / exact-synth 95 / **AQC 15** at infidelity 2.6e-4 -- i.e. AQC beats even exact synthesis by 2.1x (n=3) and 6.3x (n=4). **CONTROLLED evolution — what a Hadamard test actually runs, and where the story reverses:** against the notebook's REAL `build_controlled_evolution("exact")`, which builds the |0><0|(x)I + |1><1|(x)U block directly as one UnitaryGate rather than controlling a synthesised circuit: n=3 exact **50 CX** vs AQC 132 CX (**AQC 2.6x WORSE**); n=4 exact 236 vs AQC 216 (**AQC 1.09x, marginal**). Reason: controlling an AQC ansatz pays per-gate control overhead on every one of its gates, while the direct block construction pays none -- so AQC's raw-compression advantage is largely eaten by controlling it. Crossover sits at n~4, exactly where R025 independently found the exact-synthesis trick dying (3.41x -> 1.16x advantage over Trotter), so AQC should win beyond n=4 as O(4^n) kills the direct block construction. **METHODOLOGICAL CORRECTION made before reporting:** a first version compared AQC against a naive `.control()` of the exact-synthesis circuit (192/920 CX), which flattered AQC by ~4x and would have supported a false "AQC wins everywhere" claim; replaced with the notebook's actual construction. A second bug had the verdict logic call a 7.6x gate reduction a "loss" over a 4e-6 fidelity difference. Both fixed | infidelities 1.4e-4 / 2.6e-4; gate counts deterministic (fixed transpile seed) | 0 shots (classical compression + statevector certification); the Track-B measurement route is demonstrated in R034 at 4000 shots/quadrature | evidence/scripts/aqc_compression.py; evidence/aqc_compression_result.json; evidence/aqc_console_output.txt | 2026-08-13 R034 | **Track B (PARTIAL) — dynamics A/B tester via anti-controlled Hadamard test** | Generalises the Hadamard test: W on the |0> ancilla branch, U on the |1> branch, so the ancilla measures the OVERLAP OF TWO DYNAMICS. Derived here (this is NOT the standard circuit, so it gets its own identity and its own validation): **_phi = Re[e^{i phi}]**, i.e. chi_AB(t) = , with W=I recovering the ordinary test as a strict special case. **STATEVECTOR VALIDATION: max deviation 8.99e-15 over 3 times x 2 reps x 2 quadratures** (want <1e-10) — identity confirmed; chi_AB(0)=1.000000 exactly, as it must be since both evolutions are the identity at t=0. Study run: U = exact evolution, W = 2nd-order Suzuki-Trotter at reps in {1,2,4}, so chi_AB directly measures product-formula drift. Measured |chi_AB| (4000 shots/quadrature) vs exact, at t=0/0.9/1.8/2.7/3.6/4.5: reps=1 -> 1.000(1.000), 0.982(0.986), 0.779(0.780), 0.627(0.626), 0.596(0.620), 0.899(0.889); reps=2 -> 1.000(1.000), 1.000(0.999), 0.978(0.982), 0.789(0.766), 0.581(0.574), 0.878(0.859); reps=4 -> 1.001(1.000), 1.000(1.000), 1.000(0.999), 0.991(0.988), 0.938(0.942), 0.771(0.762). More reps stay nearer 1 at fixed t, as expected; the non-monotonic recovery at late t is genuine (Trotter error is oscillatory, not cumulative-monotonic). GATE COST: plain Hadamard (exact) depth 220 / 50 CX; A/B tester depth 866/1400/2468 and 344/588/1076 CX at reps 1/2/4 — the A/B circuit carries BOTH evolutions, matching the scaffold's own "roughly doubles the controlled-evolution depth" note. **SCOPE: PARTIAL, stated explicitly** — per-observable disagreement profiles are NOT delivered; those need the X-basis ancilla readout (rho^(X), "delete the final Hadamard"), a separate derivation not attempted | statevector identity 8.99e-15 (deterministic); overlap curves 4000 shots/quadrature/point, single seed | 6 times x 3 reps x 2 quadratures x 4000 = 144,000 shots (exploratory, not the Part A/B record) | evidence/scripts/track_b_ab_tester.py; evidence/track_b_result.json; evidence/track_b_console_output.txt | 2026-08-13 R033 | **Track E (own proposal) — ancilla-vs-system consistency as a SELF-VALIDATING decoherence witness** | Exploits the identity Tr[(rho^(I))^2] = (1+|chi|^2)/2, exact for pure input: LHS measurable from the SYSTEM register alone (shadow snapshot pairs, standard HKP per-qubit factors +5/-4/+0.5), RHS from the ANCILLA alone. Two independent channels of the same experiment -- so the gap certifies decoherence with **NO exact diagonalisation and NO classical simulation**, unlike every other check in this project, which means it remains usable at sizes where classical verification is impossible. |chi|^2 is debiased (E[mean(a)^2] = mean^2

+ Var, so the raw square is biased upward and would fake a gap even on a perfect simulator). THREE-LEG VALIDATION, all at t=0.9, balanced 3^n-basis ensembles: **(a) ideal sim -> null**: 2site gap -0.0015+-0.0130 (-0.1 sigma), frozen/n=3 gap -0.0355+-0.0272 (-1.3 sigma) -- identity confirmed, estimator unbiased. **(b) noisy sim -> fires, with dose-response**: at the notebook's own Challenge-11 toy parameters gap = -0.0794+-0.0105 (**-7.6 sigma**); at 5x those parameters gap = -0.1733+-0.0101 (**-17.2 sigma**) -- monotonic in noise strength, confirming it measures what it claims. **(c) real hardware (ibm_kingston 2site, job d9uk99k98n5s7392vhsg, R023) -> gap -0.0201+-0.0144 (-1.4 sigma), NOT significant** -- a CORRECT null, not a failure: that job is our cleanest hardware result (chi survival 0.962), so a witness reporting "no significant decoherence" is right. Note hardware (-0.020) is LESS decohered than the 1x toy model (-0.079), consistent with kingston's real 2q error 1.99e-3 being ~3x better than the toy's 6e-3 at this shallow 13-gate depth -- and NOT in tension with R014's finding that hardware is ~2x WORSE than the toy model at 435 gates, since idle/crosstalk effects absent from both models grow with depth. SECOND DELIVERABLE, same records, zero new circuits: product-state fidelity <|0..0><0..0|> expanded into 2^n Pauli strings (projector verified exactly against the dense matrix for n=2,3 before use) -- ideal 0.6330+-0.0042 vs exact 0.6337 (0.2 sigma); hardware 0.6228+-0.0060 (1.8 sigma); noisy-5x 0.5096 (30.2 sigma), degrading as expected | sigma values as tabulated; purity sem from a 20-block jackknife (see BUGLOG B06 -- the naive pair-count sem was wrong and was caught by the ideal-sim control) | ideal/noisy legs 3^n bases x 2 quadratures x 4000 shots; hardware leg reuses R023's 36,000 shots; 0 new QPU time | evidence/scripts/track_e_consistency_witness.py; evidence/track_e_result.json; evidence/track_e_console_output.txt | 2026-08-13 R032 | **Track B/C — Track C — eigenstate detective (notebook §7.5 scaffold, implemented)** | `eigenstateness_witnesses(sweep)` used verbatim from the notebook's given scaffold. Calibration family: theta=0.0 (Ry angle on q0, prep -> |000>) to 1.3 (our actual benchmark), reduced grid (16 pts, T_max=12.6 preserved, DT=0.7875), 1000 shots/setting, sem(w1) via parametric bootstrap (same style as Challenge 9). **theta=0.0 is an EXACT eigenstate** (Checkpoint 2 already establishes |000> at E=0.55) so its witness value is a genuine zero-check, not an assumption: w1=0.0016, z=w1/sem(w1)=0.42 -- correctly UNDETECTED. Monotonic, clean calibration curve as theta increases: theta=0.2 w1=0.0101 z=2.53 (borderline); 0.4 w1=0.0369 z=8.36; 0.65 w1=0.1015 z=21.94; 1.0 w1=0.2249 z=41.20; **1.3 (benchmark) w1=0.3373 z=62.30** (overwhelmingly detected). DETECTION POWER VS SHOT BUDGET (theta=0.2 weakly-mixed vs theta=1.3 our benchmark, shots in {200,800,3200}): weakly-mixed theta=0.2 needs the full 3200 shots/setting just to reach z=4.64 (still short of the project's usual 5-sigma bar); strongly-mixed theta=1.3 is already detected at z=25.42 with only 200 shots/setting, rising to z=99.86 at 3200 -- a direct, quantitative demonstration that detection power scales with HOW mixed the state is, not just with shots | z-scores as tabulated (bootstrap-derived sem, 200 replicates) | calibration: 6 theta x 16 times x 2 quadratures x 1000 shots = 192,000 shots; detection-power substudy: 2 theta x 3 shot-counts x 16 x 2 = 268,800 shots; both reduced-budget exploratory studies, NOT the official 256,000-shot Part A/B record | evidence/scripts/track_c_eigenstate_detective.py; evidence/track_c_result.json; evidence/track_c_console_output.txt | 2026-08-13 R031 | **Teammate-submitted real-hardware results (real_machine/) — 6 jobs, 2 devices incl. ibm_miami (unreachable from our own account), 3 models, all DONE. UPDATE 2026-08-14: ibm_miami IDENTIFIED AS A NIGHTHAWK-CLASS DEVICE, from our own artefacts rather than assertion — real_machine/log.txt records `ibm_miami (436 directed coupling edges)` = 218 couplers, exactly IBM Nighthawk's published square-lattice spec (120 qubits, 218 tunable couplers), and distinct from the heavy-hex Herons (`ibm_kingston (352 directed coupling edges)` = 176 couplers in the same log). So the project's hardware grid spans BOTH current IBM architectures: heavy-hex Heron (kingston, marrakesh) and square-lattice Nighthawk (miami)** | Independently run by a teammate on a different IBM account/token, using OUR unmodified hardware_run.py/layout_search.py toolchain (real_machine/log.txt shows the exact CLI invocations and console output; their build has a few extra diagnostic print lines confirming the transpiler actually honoured --initial-layout, which we adopted). Raw qiskit_ibm_runtime job dumps (job_id/backend/status/inputs/result/metrics, checked for and containing NO credentials/tokens/instance CRNs before committing) decoded with the standard RuntimeDecoder and run through the SAME graded parse_memory/estimate_hadamard_signal as every other hardware result here -- not a new analysis path. Full table (chi_hw, survival=|chi_hw|/|chi_exact|, sigma deviations re/im): frozen+kingston (job d9umb6c98n5s7393271g, searched layout 77 2q gates) chi=+0.542-0.061j, **survival 0.808**, 6.4/3.2 sigma; 2site+kingston (d9umfoou5hac73ahduog, 13 2q gates) chi=+0.377+0.078j, **survival 0.595**, 12.4/2.2 sigma; 4site+kingston (d9umbk498n5s739327k0, SAME searched window as our own pending R026 job, 386 2q gates) chi=+0.268-0.291j, **survival 0.587**, 13.4/4.2 sigma; frozen+miami (d9umcld35hes73fk4g0g, 82 2q gates as submitted; layout_search's searched-window count was 80) chi=+0.498-0.014j, **survival 0.738**, 8.5/5.3 sigma; 2site+miami (d9umg0gu5hac73ahdv10, 13 2q gates) chi=+0.537-0.347j, **survival 0.988**, 5.2/**22.6** sigma; 4site+miami (d9umcab43mgs73et5th0, 370 2q gates) chi=+0.201+0.021j, **survival 0.300**, 16.2/17.9 sigma. TWO GENUINE FINDINGS, neither smoothed over: (1) **survival (magnitude ratio) can hide a large phase error** -- 2site+miami has near-perfect survival (0.988) yet its imaginary component is 22.6 sigma off (exact +0.1277, measured -0.347 -- opposite-ish sign despite matching |chi|), so survival alone is NOT sufficient to certify a result; both re/im deviations must be checked. (2) **naive per-2q-gate-error survival prediction is unreliable across configurations**: miami/2site predicted 0.9875 (layout_search) vs measured 0.988 (near-perfect); kingston/2site predicted 0.987 vs measured 0.595 (large miss, same prediction methodology, same device family) -- readout error, T1/T2 idle decoherence, and calibration drift since the error snapshot are not captured by the simple per-edge product model, and its reliability varies configuration to configuration rather than being a fixed correction factor. Also notable: kingston/frozen's SEARCHED layout (77 gates, predicted 0.9187) measured survival 0.808, LOWER than our own AUTO-layout job's 0.878 (R018, 101 gates) -- fewer predicted-optimal gates did not beat the naive default here, reinforcing that single-run hardware comparisons carry real device-noise-day variance and the searched layout is not a guaranteed win per run | n/a (6 single runs; sigma deviations as tabulated per job, no repeated-trial averaging) | 6 jobs x 2 circuits x 2000 shots = 24,000 shots, run on the teammate's account/quota, not ours | evidence/scripts/load_real_machine_results.py; evidence/real_machine_analysis.json; raw dumps in real_machine/*.json; real_machine/log.txt (teammate's console transcript) | 2026-08-13 R030 | Ablation #4 — standard Quantum Phase Estimation (the item never built from the original 5-item ablation request) | Design: m=5 counting qubits, t0=T_max/2^m=0.39375 -- chosen so QPE's total evolved time (2^m-1)*t0=12.21 matches the SAME budget (T_max=12.6) the DFT/pencil analysis gets, and its energy bin width 2*pi/(2^m*t0)=0.4987 EXACTLY equals the DFT's Rayleigh resolution 2*pi/T_max=0.4987 -- neither method given an artificial resolution advantage. No-aliasing check: pi/t0=7.98 vs full spectral radius 2.635, comfortably clear. controlled-U(2^k*t0) built directly via the graded build_controlled_evolution at each required time (method="exact"), not by repeating a base circuit -- reuses already-validated code, standard efficient QPE-simulation practice. RESULT (50,000 shots, ideal simulator): **4/4 populated lines recovered**, max|dE|=0.0794 -- better than DFT's 3/4 (R011, misses the -0.481 line entirely) but worse than the pencil's 4/4 at max|dE|=0.0325 (R012). QPE avoids DFT's windowing-leakage failure mode (direct projective phase measurement, not a windowed correlation) while still being grid-limited (discrete counting-register outcomes) rather than the pencil's continuous off-grid fit -- QPE genuinely sits BETWEEN the two, exactly the intermediate behaviour the design predicts. **LABELS: NONE, at any cost** -- QPE's ancilla qubits are entirely consumed by phase estimation; there is no spare "garbage register" to shadow-read, unlike the Hadamard-test protocol where the SAME physical resource (the ancilla + system qubits already being measured) yields the symmetry label for free. This is the clearest statement of what the shadow method actually buys: not better energies (pencil beats both), but labels from no additional measurement resource, which QPE structurally cannot provide regardless of engineering effort. COST: 8 total qubits (3 system + 5 counting, vs 4 for the shadow-Hadamard test at the same system size) and **277 CX / depth 1081** for the single circuit. LIKE-FOR-LIKE COMPARISON (C4 audit correction): this 277 is transpiled to `basis_gates` with NO coupling map, so it must be compared against the shadow-Hadamard exact path's all-to-all count of **50 CX** (R034), i.e. QPE is **5.5x deeper** -- NOT against R018's 101 CX, which is coupling-mapped to kingston. An earlier version of this row also called 277 'comparable to its Trotter path (435 CX)'; that was likewise mixing bases, since the all-to-all Trotter count is 68 CX (R035). The qualitative conclusion (QPE is the most expensive method here) survives and strengthens | n/a (single ideal-sim run, deterministic given seed) | 50,000 shots, single circuit design (no basis variation needed) | evidence/scripts/qpe_ablation.py; evidence/qpe_ablation_result.json | 2026-08-13 R027 | Going-further Track A item 1 — rank selection, replacing the sv_threshold=0.06 magic number | Singular values of the Hankel matrix (first 10 of 32): 20.113, 5.263, 4.605, 1.654, 0.327, 0.277, 0.262, 0.223, 0.198, 0.181. Principled criterion derived from the MEASURED per-point sem (noise floor = sigma*(sqrt(L)+sqrt(M)) for an LxM matrix of iid noise, sigma=0.01958): floor=0.2215, giving rank 8 — MORE PERMISSIVE than the true rank 4, and than sv_threshold=0.06's rank 4. HONEST RESULT: the "principled" derivation does NOT beat the magic number here; threshold sweep (9 points, 0.005-0.40) shows *sv_threshold in [0.02,0.06] all give the correct rank=4/4 modes/4 labels, while the derived noise-floor criterion would over-include 4 low-SNR singular vectors. IMPORTANT QUALIFIER (C4 audit): over-including them is NOT harmful here -- the sweep shows rank 8 (thresh 0.01) still yields 4 modes, 4/4 labels and max|dE|=0.0321, marginally BETTER than rank 4's 0.0325. So the principled criterion fails to beat the magic number, but it does not degrade the result either. Reported as-is, not spun as a win | n/a (deterministic given the fixed record set) | reuses R009's 256,000 shots; 0 new shots | evidence/scripts/going_further_trio.py section (1) | 2026-08-13 R028 | Going-further cell-62 item 4 / Track A item 2 — measured chi/chi_Q correlation + correlated bootstrap | corr(a, a*Qhat) measured DIRECTLY from our own shot records (not assumed): phi=0 mean +0.534 (sd 0.114, 1 of 64 deterministic t=0 points excluded via nanmean — c-U(0)=I collapses the ancilla exactly there); phi=-pi/2 mean +0.605 (sd 0.050). Consistent with the notebook's quoted ~+0.57. CORRELATED bootstrap (200 replicates, draws chi/chi_Q jointly at the measured per-time correlation instead of independently) vs a fresh 200-replicate INDEPENDENT bootstrap run inside the same script (q_sd 0.0601/0.0595/0.0190/0.1826 -- not R013's own values, which came from a different seed; the ratio is internally consistent because both legs are computed together): q_sd shrinks to 0.78/0.90/0.72/0.78 of the independent value per line (mean shrink 0.80, i.e. **~20% tighter label uncertainty**); E_sd changes are smaller and mixed (0.97/1.01/1.18/1.16). CONFIRMS the notebook's claim that ignoring the correlation makes the existing bootstrap conservative, quantified: ~20% on q, not on E | n/a (point estimate of a measured correlation + bootstrap spread) | reuses R009's 256,000 shots; bootstrap is pure post-processing, 400 reconstruct() calls (200 indep + 200 correlated), 0 new shots | evidence/scripts/going_further_trio.py section (2) | 2026-08-13 R029 | Going-further cell-62 item 2 — variance budgeting, 3^w rule vs measured, term by term | Predicted sem from E[Phat_i Phat_j] via the 3^w rule (same-term: 3^w(P); cross-term: exact via matrix product, not assumed independent), N=256,000 pooled shots: sem(<H>) predicted 0.00800 vs measured 0.00817 (ratio **1.021**); sem(<Q>) predicted 0.00520 vs measured 0.00520 (ratio **0.999**). Both within 2% — the 3^w variance model quantitatively explains the measured error bars, not just their order of magnitude | ratios 1.021 / 0.999 | reuses R009's 256,000 shots (pooled over ALL_RECS, matching R010's H_HAT/Q_HAT/H_SEM/Q_SEM exactly); 0 new shots | evidence/scripts/going_further_trio.py section (3) | 2026-08-13"

R042 | **Track B COMPLETED — the per-observable disagreement profile R034 left undone, plus 2-site validation and hardware sizing** | The scaffold's missing piece is the X-basis ancilla readout ("delete the final Hadamard"). Derived here: before the final Hadamard the state is (|0>W|psi> + e^{i phi}|1>U|psi>)/sqrt2, so tr_a[rho_out] = (W rho W+ + U rho U+)/2 and tr_a[(Z x I) rho_out] = (W rho W+ - U rho U+)/2 — BOTH phi-INDEPENDENT, so one quadrature suffices, and sum+difference give **_W and _U SEPARATELY** for every Pauli O from one circuit family. **VALIDATION BEFORE ANY QPU TIME: arm A (overlap) identity max dev 4.56e-15 over 4 times x 2 reps x 2 quadratures; arm B (profile) identity max dev 4.44e-15 over 4 times x 6 observables. BUG CAUGHT BY THIS CHECK (endianness class, CONVENTIONS-flagged):** the first reference extracted the |1>-control block as the UPPER HALF of the gate matrix, but build_controlled_evolution documents 'qargs = [anc, s0, ...] -> ancilla is the LOWEST matrix bit', so the block is the odd sublattice [1::2,1::2]. The wrong version passes at t=0 (both branches identity) and fails at every t>0 — worst dev 7.04e-01. Fixed before submission. HARDWARE SIZING on ibm_kingston: ordinary test 13 two-qubit gates / depth 54 / predicted survival 0.984 (sanity: R023 MEASURED 0.962 at 15 gates, so the model runs ~2% optimistic); A/B reps=1 **136 gates / depth 533 / predicted 0.845**, t/T1 = 0.082; A/B reps=2 235 gates / depth 927 / predicted 0.783 | identities deterministic (statevector); survival predictions are gate-error-only upper bounds, stated as such | 0 shots (validation + transpile only) | evidence/scripts/track_b_hardware_prep.py; evidence/scripts/track_b_hardware_submit.py --dry-run; evidence/track_b_hw_sizing.json | 2026-08-13 R043 | **Track B measured against the baselines the compiling literature actually uses — and it loses on cost** | Comparing our arm only against 'discard the garbage' would be a strawman, so the real baselines were implemented and VERIFIED against exact values first: **Loschmidt echo P(0..0) = ||^2, dev 3.66e-15; Hilbert-Schmidt test (Khatri, LaRose, Poremba, Cincio, Sornborger, Coles, Quantum 3, 140 (2019)) P(0..0) = |Tr[W+U]/d|^2, dev 6.66e-16.** TRANSPILED COST on ibm_marrakesh, 2-site, t=0.9, W=Trotter(reps=1): classical fidelity 0 qubits/0 circuits (NOT SCALABLE — needs simulation); **Loschmidt echo 2 qubits, 1 circuit, 2 two-qubit gates, depth 17**; Hilbert-Schmidt 4 qubits, 1 circuit, 16 gates, depth 43; Track B ancilla-only 3 qubits, 2 circuits, 136 gates, depth 536; Track B + shadows 3

qubits, 27 circuits, same 136 gates. **SCALING CHECK, because the n=2 gap is an artefact:** at n=2 any product of 2-qubit unitaries fuses into one, so the transpiler collapses the echo. All-to-all, optimization_level=3: n=2 echo 2 vs Track B 105 (52.5x); n=3 39 vs 235 (6.0x); n=4 127 vs 510 (4.0x). **CONCLUSION STATED AGAINST OUR OWN METHOD: the echo is cheaper at every n. Quote 4-6x, never 52x.** Track B's justification is what it RETURNS — the phase (the echo and HST give magnitude only) and the per-observable profile (neither gives it) — never cost. The HST answers a STRONGER, state-independent question at the price of 2n qubits | baseline identities deterministic; gate counts deterministic (fixed transpile seed) | 0 shots (classical) | evidence/scripts/track_b_baselines.py; evidence/track_b_baselines.json | 2026-08-13
R044 | **TRACK B ON A QPU — first hardware run of Track B in this project, with two literature baselines in the SAME job** | Prior hardware rows are all Part A (R008 frozen, R023 2-site, R026 4-site, R031 teammate) or reuse R023's shots (R033/R036/R037); Track B and Track C had never run on a QPU. Four arms submitted together so the comparison is same-device, same-hour: A = chi_AB from the ancilla (the GARBAGE baseline), B = _W and _U separately from shadows (ours), C = Loschmidt echo, D = Hilbert-Schmidt. 2-site side model (ROBUSTNESS CLAIMS ONLY), W = 2nd-order Trotter reps=1, U = exact, 116 circuits x 1000 shots = 116,000 shots, all 3^2 = 9 bases (full balanced ensemble, so the shadow estimator is valid — the B04 condition). **ibm_marrakesh job `d9uss4f2sl0c73bl4em0` RETURNED.** Loschmidt echo P0 = 1.0000 / 0.9650 / 0.5420 / 0.1340 vs exact 1.0000 / 0.9738 / 0.5854 / 0.1069; Hilbert-Schmidt P0 = 0.9970 / 0.8790 / 0.2900 / 0.0120 vs exact 1.0000 / 0.9640 / 0.3161 / 0.0063; Track B |chi_AB| = 0.9939 / 0.5706 / 0.6304 / 0.4413 vs exact 1.0000 / 0.9868 / 0.7651 / 0.3270 (t = 0.0/0.9/1.8/2.7). **THE RESULT GOES AGAINST US AND IS REPORTED AS SUCH: the 2-gate echo tracks the exact curve to ~0.04 throughout, while the 136-gate Track-B arm is clean only at t=0 (arm-B per-observable errors all <=0.053) and is noise-dominated from t=0.9 on.** The predicted 0.815-0.845 survival was too optimistic; the t=2.7 point came back ABOVE 1, which is unphysical for pure damping and shows the errors are not simple depolarising loss. **SYMMETRY CHANNEL AS A REFERENCE-FREE ERROR BAR (the one clean win):** [H,Q]=0 and every Trotter factor conserves Q, so “_W - ‘_U is EXACTLY zero for any W in this family; measured -0.044 / -0.393 / -0.251 / -0.335 at t = 0.0/0.9/1.8/2.7. That deviation is device error certified without any reference state, exact diagonalisation or simulation, and it flags precisely the times where the other observables degraded. Diagnostic only — NOT post-selected on, which would bias the shadow estimators (same failure class as B04). **KINGSTON LEG RETURNED (2026-08-13), and it confirms the finding on a second device.** job `d9uss5n2sl0c73bl4eog`: echo P0 = 0.9980 / 0.9580 / 0.5800 / 0.1410; HST P0 = 0.9900 / 0.8630 / 0.2700 / 0.0110; |chi_AB| = 0.9989 / 0.5930 / 0.4506 / 0.0841; ‘_W - ‘_U = +0.007 / -0.435 / +0.529 / -0.103. **WORST ERROR vs exact ACROSS BOTH DEVICES AND ALL FOUR TIMES: Loschmidt echo 0.0434, Hilbert-Schmidt 0.1010, our Track B arm 1.1879 — a 27x gap in the baselines' favour.** NEW INFORMATION FROM THE SECOND DEVICE: on kingston the Track B arm fails in PHASE, not merely in magnitude — at t=0.9 it returns -0.0336-0.5920j against exact +0.9867+0.0164j, i.e. rotated by roughly 90 degrees rather than damped toward zero. Together with the marrakesh t=2.7 point coming back ABOVE unit survival, this rules out a simple depolarising-damping description of the 136-gate circuit's failure. The echo held on both devices (worst 0.043), so the gap is attributable to circuit depth rather than to either device being unusually bad. | single run per configuration, single seed; no bootstrap (hardware, exploratory) | 116,000 shots per backend | evidence/scripts/track_b_hardware_submit.py ibm_marrakesh ibm_kingston; evidence/scripts/track_b_hardware_fetch.py; evidence/track_b_hw_result.json | 2026-08-13’’’”

R045 | **PROOF OF CONCEPT (future work, NOT a Topic-5 deliverable) — the Track B circuit IS a density-of-states estimator, linking it to Goh & Koczor arXiv:2407.03414v2** | Track B's ancilla measures chi_AB(t) = Tr[W(t)^dag U(t) rho]. Setting rho = I/d makes it STATE-INDEPENDENT: chi_AB = Tr[W^dag U]/d, and at W=I it is Tr[U(t)]/d — exactly the generating function behind that paper's Eq. (7), G(t) = (1/sqrt(2pi)) Tr[e^{-iHt}], whose Fourier transform is the DOS. **So their DOS protocol is the W=I special case of our Track B circuit, not a separate experiment.** This sharpens R038, which showed the shadow-Hadamard PIPELINE transfers; this shows the TRACK B CIRCUIT contains it. Mixed input realised by enumerating all d = 2^n computational basis states (Tr[A] = sum_z), the deterministic version of their Statement 2 (a 1-design, e.g. random {I,X}^n, suffices). **STATEVECTOR VALIDATION: max deviation 2.22e-15 over 2 W-choices x 4 times** (want <1e-10). **SHOT MEASUREMENT (AerSimulator, 4000 shots/setting, 13 times x 2 quadratures x 4 basis states): Tr[U]/d rms error 0.0085; Tr[W^dag U]/d rms error 0.0072**; both give chi(0) = 1.0000 as they must. **THE COMPARISON THAT MOTIVATES IT:** our Hilbert-Schmidt arm (R043/R044) measures the SAME process overlap but returns |Tr[W^dag U]/d|^2 on 2n qubits; this route returns the COMPLEX value on n+1 qubits. **BUT THE PHASE ADVANTAGE IS CASE-SPECIFIC, AND WE CHECKED RATHER THAN ASSUMED: max|Im Tr[U]/d| = 0.5040 (the DOS case — the HST genuinely loses information), while max|Im Tr[W^dag U]/d| = 2.8e-14, i.e. ZERO to machine precision for the Trotter case.** Reason: real symmetric H gives U = *U^dag, and a SYMMETRIC (palindromic) Suzuki sequence gives W* = W^dag, which forces Tr[W^dag U] real. So against the HST our edge is the full complex value for DOS, and only register size (n+1 vs 2n) for Trotter. A first draft of this row claimed the phase advantage unconditionally and cited a t=1.8 example whose phase was -0.0000 rad; corrected before reporting. **BONUS DIAGNOSTIC:** Im Tr[W^dag U] = 0 is an identity for this class, so a measured nonzero imaginary part is pure device error — the same reference-free indicator as the conserved-charge channel (R044), on a different observable. **SCOPE, STATED PLAINLY:** proof of concept only. The mixed input is built by exhaustive enumeration, which is affordable solely because d=4 here; the 1-design sampling error at scale is NOT characterised, the Q^k moment inversion sketched in R038 is still NOT implemented, and no advantage over classical computation is claimed | statevector identity deterministic (2.22e-15); shot rms as tabulated, single seed, no bootstrap (prototype) | 13 times x 2 quadratures x 4 basis states x 2 W-choices x 4000 = 832,000 shots, simulator only, 0 QPU time | evidence/scripts/track_b_dos.py; evidence/track_b_dos.json | 2026-08-13

R046 | **AQC MAKES THE HADAMARD TEST SCALE — crossover measured at n=4, 35.9x fewer two-qubit gates by n=7 — AND a phase trap that silently destroys the answer, fixed with one gate** | R035 left this open with two points (AQC 2.6x WORSE at n=3, 1.09x better at n=4) and an EXTRAPOLATION that it 'should win beyond n=4'. Measured here at n=3..7 against the honest baseline (the notebook's own `build_controlled_evolution("exact")`, which builds |0><0|(x)I + |1><1|(x)U directly as one UnitaryGate — markedly cheaper than a naive .control() of a synthesised circuit, and therefore harder to beat). **CONTROLLED two-qubit gate counts: n=3: exact 50 / Trotter-r2 538 / AQC 132 CX (0.38x); n=4: exact 236 / Trotter-r2 818 / AQC 216 CX (1.09x); n=5: exact 1004 / Trotter-r2 1098 / AQC 300 CX (3.35x); n=6: exact 4140 / Trotter-r2 1378 / AQC 384 CX (10.78x); n=7: exact 16812 / Trotter-r2 1658 / AQC 468 CX (35.92x).** The exact block grows like 4^(n+1) (50 -> 236 -> 1004 -> 4140 -> 16812) while controlled AQC grows roughly linearly (132 -> 216 -> 300 -> 384 -> 468), so the crossover is STRUCTURAL, not a tuning artefact — though measured only to n=7. AQC state infidelity is flat at 1.4e-4..3.0e-4 throughout. **THE TRAP, found by this row and not previously known to us: qiskit-addon-aqc-tensor optimises STATE FIDELITY ||, which is BLIND TO GLOBAL PHASE — and a Hadamard test is precisely an interferometer that measures global phase.** A compression converged to state infidelity 3e-4 therefore yields chi = wrong by 2.3-3.0 RADIANS. **|chi error| WITHOUT correction, n=3..7: 1.350 / 1.224 / 1.291 / 1.330 / 1.313 — on a quantity bounded by |chi|<=1, i.e. the error is as large as the signal and the test is worthless.** R035's own certificate used |chi_AB|, a MAGNITUDE, so it could not have detected this; that is a real gap in R035 now closed. **THE FIX COSTS ONE SINGLE-QUBIT GATE:** if W|psi> = e^{i theta} U|psi> then theta = arg() is available at compile time (the same MPS run that produces W gives it free), and a P(-theta) on the ANCILLA cancels it exactly, because the |1> branch is the one carrying W. **|chi error| WITH the correction: 0.0100 / 0.0080 / 0.0082 / 0.0079 / 0.0083 — a 135x to 168x improvement. SCOPE, STATED PLAINLY AND NOT BURIED:** AQC here is compressed against ONE input state, so its PROCESS infidelity is large (0.48 / 0.67 / 0.97 / 0.96 / 0.96 for n=3..7) — this is NOT a general-purpose controlled-U and must not be reused with a different input. It is legitimate for a Hadamard test, whose controlled gate only ever acts on that one prepared state, and illegitimate the moment anything feeds it another. Both state and process infidelity are reported side by side in the script output. Measured to n=7 only; no claim beyond. No advantage over classical computation is claimed — these are gate counts, not runtimes | gate counts deterministic (fixed transpile seed, optimization_level=1); infidelities and chi errors from exact statevector, no shot noise | 0 shots (classical compilation study) | `python evidence/scripts/aqc_scaling.py 3 4 5 6 7`; evidence/aqc_scaling.json; chart evidence/scripts/make_aqc_chart.py | 2026-08-14

R047 | **SKQD's scaling advantage TESTED at a size where it could appear — and it does NOT, for our Hamiltonian. Negative result with a measured mechanism and a boundary** | R037 stated plainly that it could not test SKQD's value proposition (spanning a subspace too large to enumerate) because our benchmark's reachable space is only 3-4 dimensional. This row builds a sector where the test is meaningful: the chain conserves Q, so preparing a HALF-FILLED state gives a sector of dimension C(n,n/2) — 252 (n=10), 924 (n=12), 3432 (n=14). **Sector Hamiltonian validated against the full 2^n matrix first: max|H_sector - H_full[sector]| = 0.00e+00 at n=4 and n=6, ground energies identical to 8 decimals.** Protocol is SQD's actual one: sample configurations from time-evolved states, RANK by observed frequency, diagonalise H in the span of the top M. **PREMISE HOLDS AND IMPROVES WITH n:** 90% of the ground state's weight lives on n=10: 61/252 = 24.2%; n=12: 172/924 = 18.6%; n=14: 484/3432 = 14.1% — a SHRINKING fraction, which is exactly what SKQD assumes. **BUT THE ADVANTAGE DOES NOT MATERIALISE:** subspace needed to reach 1% relative error — n=10 (sector 252): needs 200/252 = 79.4%; n=12 (sector 924): needs 800/924 = 86.6%; n=14 (sector 3432): needs 3200/3432 = 93.2%. **The fraction RISES with n (79.4% -> 86.6% -> 93.2%), the opposite of the intended demonstration.** So the ground state is concentrated, but sampling from a time-evolved reference does not RANK the important configurations first; the gap between those two facts is where SKQD lives or dies. **MECHANISM IDENTIFIED, by a control that sweeps Delta/J at n=12 (all else fixed): Delta/J=0.38: 90% on 19%, needs 800/924 = 86.6%; Delta/J=1.00: 90% on 11%, needs 400/924 = 43.3%; Delta/J=2.00: 90% on 3%, needs 150/924 = 16.2%; Delta/J=5.00: 90% on 0%, needs 25/924 = 2.7%; Delta/J=10.00: 90% on 0%, needs 25/924 = 2.7%.** SKQD works superbly when the ground state is LOCALISED (Delta/J >= 5 needs 2.7% of the sector) and fails when it is DELOCALISED. **Our frozen benchmark sits at Delta/J = 0.25/0.65 = 0.38, hopping-dominated — the worst case.** This is therefore a property of the Hamiltonian we were given, NOT a refutation of SKQD, which is normally applied to chemistry problems where a Hartree-Fock reference dominates the ground state (the localised end of this same axis). **A FALSE START, recorded rather than hidden:** the first version drew a fixed shot budget and unioned every observed configuration, which measures the sampler's coverage rather than the method, needed ~99% of the sector, and looked like a much worse failure than it is. Replaced with the top-M-by-frequency protocol above. A second check — whether a better reference rescues it (Neel vs the lowest-energy configuration of H's diagonal part, the spin-chain analogue of Hartree-Fock) — was inconclusive: identical at n=10,12 (the two references coincide) and the n=14 point improved while having WORSE ground overlap, i.e. single-seed sampling noise, so no claim is made from it. **RECOMMENDATION: do not headline SKQD.** The honest deliverable here is the boundary, not a win | deterministic sector Hamiltonian (validated 0.00e+00); convergence curves single-seed (RNG 2026), no bootstrap — the qualitative trend is monotone across 3 sizes and 5 Delta/J values | 0 shots (classical; sampling is simulated from exact statevectors) | `python evidence/scripts/skqd_scaling.py`; evidence/skqd_scaling.json | 2026-08-14

R048 | **The SKQD boundary in 2D: connectivity hurts only where hopping already wins — same Hilbert space, different geometry** | Follow-up to R047, which located SKQD's boundary in Delta/J on a 1D chain. Geometry is the other knob on delocalisation, so this compares a **1D open chain (12 sites, 11 bonds)** against a **2D 3x4 grid (12 sites, 17 bonds)**. Same site count, same conserved charge, therefore **the same sector dimension C(12,6) = 924** — any difference is connectivity alone, not Hilbert-space size. Same top-M-by-frequency protocol, same 1% relative-energy target. **RESULT: Delta/J=0.38: 1D 86.6% (ground on 19%) vs 2D NEVER (>86.6%) (ground on 44%); Delta/J=1.0: 1D 64.9% (ground on 11%) vs 2D 86.6% (ground on 22%); Delta/J=2.0: 1D 16.2% (ground on 3%) vs 2D 16.2% (ground on 1%); Delta/J=5.0: 1D 2.7% (ground on 0%) vs 2D 2.7% (ground on 0%). So the two regimes behave completely differently.** In the DELOCALISED regime (Delta/J <= 1) more bonds make SKQD strictly worse — at our benchmark's Delta/J = 0.38 the 2D ground state spreads over 44% of the sector against 1D's 19%, and 2D never reaches 1% error within the budget at all. In the LOCALISED regime (Delta/J >= 2) the two geometries are **identical** (16.2% and 2.7% in both), because once the ground state collapses onto a handful of configurations it no longer matters how they are connected. **CONSEQUENCE FOR US: a 2D version of our benchmark would be worse for SKQD, not better.** Delta/J is the governing parameter; geometry amplifies it in one regime and is irrelevant in the other. **A BUG THIS CAUGHT, recorded because it inverted the headline:** the first version hard-coded a 1D Neel reference (alternating sites 0,2,4,...) onto the 2D grid, where that pattern is NOT the antiferromagnetic configuration. The sampler therefore never visited the true ground configuration and 2D appeared to fail even at Delta/J=5, where the ground state is a SINGLE configuration — an impossible result that exposed the error. Replaced with the lowest-diagonal configuration of H itself (the classical Ising ground state on whatever bond list is passed), which is geometry-agnostic, costs no diagonalisation, and is the spin-chain analogue of a Hartree-Fock reference | single seed (RNG 2026), no bootstrap; trends are monotone across 4 Delta/J values and consistent between geometries in the localised regime | 0 shots (classical; sampling simulated from exact statevectors) | `python evidence/scripts/skqd_geometry.py`; evidence/skqd_geometry.json; chart evidence/scripts/make_skqd_chart.py | 2026-08-14

R049 | **The AQC headline SURVIVES on a 2D lattice — crossover shifts one step right, advantage costs ~18%, and MPS does not break down** | R046's 36x result was measured on a 1D open chain, and AQC-Tensor is an MPS method, native to 1D. Three reasons it could have failed in 2D were stated before running: (a) MPS on a 2D grid mapped to a linear qubit order needs bond dimension growing with the cut perimeter; (b) a grid has more bonds at equal site count (4/7/10 vs 3/5/7 at n=4/6/8), so every Trotter-derived circuit including the AQC seed gets longer; (c) the exact controlled block is GEOMETRY-BLIND (it synthesises an arbitrary (n+1)-qubit unitary at ~4^(n+1)), so 2D lengthens our side of the comparison and not the baseline's. **MEASURED (controlled two-qubit gates, 1D chain vs 2D grid at matched n): n=4: 1D 1.09x vs 2D 0.92x (exact 236 both, AQC 216 vs 256 CX, 2D infid 5.7e-05); n=6: 1D 10.78x vs 2D 8.92x (exact 4140 both, AQC 384 vs 464 CX, 2D infid 1.1e-04); n=8: 1D 122.75x vs 2D 100.83x (exact 67756 both, AQC 552 vs 672 CX, 2D infid 2.0e-04). INTERNAL CONSISTENCY CHECK PASSED:** the exact-block count is IDENTICAL across geometries at every n (236 / 4140 / 67756), exactly as predicted from its geometry-blindness — a wrong bond list would have shown up here. **VERDICT: the crossover moves right by one step (n=4 in 1D, n~5 in 2D — at n=4 the 2D case is a 0.92x LOSS) and the advantage at n=8 falls from 122.8x to 100.8x, i.e. an 18% cost for doubling the connectivity. It does not break.** Contrary to concern (a), AQC state infidelity was actually BETTER in 2D at these sizes (5.7e-5 to 2.0e-4, against 2.6e-4 to 3.0e-4 in 1D): at n<=8 the 2x4 grid's MPS bond dimension is still modest, so the 1D-native limitation had not yet bitten. **We do NOT extrapolate past n=8 — the MPS concern is deferred, not refuted.** The phase trap and its one-gate fix (R046) persist unchanged in 2D: |dchi| after correction 0.0049-0.0065, comparable to 1D's 0.008. **TRACK B BASELINES on the same geometries: n=4: echo/TrackB 5.2x (1D) vs 5.7x (2D); n=6: echo/TrackB 2.6x (1D) vs 2.8x (2D); n=8: echo/TrackB 2.3x (1D) vs 2.3x (2D).** The Loschmidt echo's cost advantage over Track B is essentially geometry-independent, so 2D does not rescue our arm. It does however SHRINK with n (5.2x -> 2.3x in 1D), because at larger n the echo is itself dominated by exact-synthesising U — the same 4^n bottleneck. The echo's dramatic cheapness reported at n=2 in R043 (2 CX) is therefore a small-n effect, and R043's own scaling note already flagged that direction | gate counts deterministic (fixed transpile seed, optimization_level=1); infidelities and chi errors from exact statevector, no shot noise; single AQC optimisation per cell (L-BFGS-B, maxiter 400) | 0 shots (classical compilation study) | `python evidence/scripts/geometry_2d.py`; evidence/geometry_2d.json | 2026-08-14

R050 | **The SKQD localisation boundary REPRODUCES on the 2D Fermi-Hubbard model — our Delta/J finding is physics, not an artefact of the XXZ benchmark** | R047/R048 located SKQD's boundary in Delta/J on the spin chain we were handed. The canonical model for SQD/SKQD is Fermi-Hubbard, whose analogous knob is U/t (large U/t = Mott = localised, small = metallic = delocalised). If our boundary is real, it must reappear there. **No qiskit-nature on this machine, so the Jordan-Wigner mapping was written by hand and validated on FOUR independent checks first, each of which a plausible bug would break: [H,N_up] = [H,N_down] = 0.0e+00 exactly; at U=0 the full 2^(2L) spectrum matches the free-fermion many-body spectrum (all occupation subsets of the single-particle levels) to 1.9e-15 and 8.4e-15; at t=0 every eigenvalue is an exact integer multiple of U (deviation 0.0e+00). Verified on both a 1x3 chain and a 2x2 grid. RESULT (2x3 grid, 6 sites / 12 qubits, half filling N_up=N_down=3, sector dimension 400, accuracy target 1% of the sector BANDWIDTH): U/t=0.5: 90% of ground on 60%, NEVER (all 278 observed configurations insufficient); U/t=2: 90% of ground on 50%, NEVER (all 398 observed configurations insufficient); U/t=4: 90% of ground on 33%, NEVER (all 395 observed configurations insufficient); U/t=8: 90% of ground on 11%, needs 150/400 = 37.5%; U/t=16: 90% of ground on 4%, needs 50/400 = 12.5%. The boundary is the same one: SKQD works in the Mott regime (12.5% of the sector at U/t=16, 37.5% at U/t=8) and fails in the metallic regime (U/t <= 4, where even every configuration the sampler observed is insufficient).** This is why SQD succeeds in the chemistry literature —

those systems sit at the localised end of exactly this axis — and why it does nothing for our hopping-dominated benchmark. **A METRIC BUG CAUGHT HERE, worth recording because it faked a total failure:** the first run used error relative to |E_0| and reported 'never converged' at EVERY U/t including U/t=16, where the ground state occupies just 4% of the sector. At half filling and large U/t the Hubbard ground energy tends to zero (~ -t^2/U), so a relative-to-|E_0| threshold tightens without bound and measures the metric rather than the method. Rescaled to the sector bandwidth, which is comparable across U/t | single seed (RNG 2026), no bootstrap; monotone across 5 U/t values | 0 shots (classical; sampling simulated from exact statevectors) | `python evidence/scripts/hubbard_2d.py`; evidence/hubbard_2d.json | 2026-08-14 R051 | **The AQC crossover ALSO survives on Fermi-Hubbard — 34.7x at 8 qubits — but the compression accuracy degrades ~30x, and that cost is NOT negligible** | R046/R049 measured the AQC advantage on XXZ chains and grids. Hubbard is the harder test: Jordan-Wigner strings make every hopping term non-local, so the Trotter circuit and the AQC ansatz seeded from it are both far longer, while the exact controlled block is unchanged at ~4^(nq+1) — both effects work against us. Run at U/t = 4, the standard correlated point. **RESULT: 1x2 (4q): exact 236 vs AQC 256 CX = 0.92x, infidelity 1.4e-03, |dchi| after phase fix 0.0213; 1x3 (6q): exact 4140 vs AQC 464 CX = 8.92x, infidelity 1.1e-02, |dchi| after phase fix 0.0353; 2x2 (8q): exact 67756 vs AQC 1952 CX = 34.71x, infidelity 1.0e-02, |dchi| after phase fix 0.0459.** Crossover at 6 qubits (a 3-site chain), against 4 qubits on the XXZ chain, and 34.7x at 8 qubits (2x2 grid, 4 sites). **THE GATE-COUNT WIN SURVIVES. THE COST DOES NOT SURVIVE UNCHANGED, and this is the part that must not be dropped:** AQC state infidelity is 1.0e-2 on Hubbard against 2.6e-4 to 3.0e-4 on the XXZ chain — roughly 30x worse — and the residual chi error after the ancilla phase correction rises from 0.005-0.008 to **0.021-0.046**. At 2000 shots the shot-noise sem on chi is ~0.02, so on Hubbard the AQC systematic error is AT OR ABOVE the statistical error and would not average away. **The honest reading: on Hubbard the compression is cheap but not yet accurate enough to use as-is; it would need a deeper ansatz or a longer optimisation, which would eat into the 34.7x.** We did not tune it further and do not claim it is usable there | gate counts deterministic (fixed transpile seed); infidelities and chi errors from exact statevector, no shot noise; single L-BFGS-B run per cell (maxiter 400) | 0 shots (classical compilation study) | `python evidence/scripts/hubbard_2d.py`; evidence/hubbard_2d.json | 2026-08-14

R052 | **The AQC advantage SURVIVES real device connectivity — and in fact WIDENS, including one case that flips from a loss to a win** | R046/R049/R051 counted two-qubit gates with all-to-all connectivity (basis gates only, no coupling map). Real IBM devices are heavy-hex, and a 2D lattice needs SWAP routing a 1D chain does not, so the stated concern was that our 2D advantage is an artefact of ignoring routing. **Re-transpiled every circuit against the actual `ibm_marrakesh` coupling map and basis (optimization_level=1, seed 2026). No QPU time — this is compilation only. RESULT: n=4 1D: exact 236->476 CX, AQC 216->324 CX, ratio 1.09x -> 1.47x; n=4 2D 2x2: exact 236->476 CX, AQC 256->373 CX, ratio 0.92x -> 1.28x; n=6 1D: exact 4140->8850 CX, AQC 384->576 CX, ratio 10.78x -> 15.36x; n=6 2D 2x3: exact 4140->8850 CX, AQC 464->701 CX, ratio 8.92x -> 12.62x. Routing widens the gap in every cell, for a structural reason:** the exact controlled block is a DENSE (n+1)-qubit unitary whose synthesis assumes all-to-all interaction, so heavy-hex routing costs it ~2.1x (236->476, 4140->8850); the AQC ansatz is shallow and mostly local, so it costs only ~1.5x (216->324, 384->576). **Most notably, n=4 on the 2x2 grid flips from a 0.92x LOSS (ideal) to a 1.28x WIN (routed)** — i.e. R049's statement that the 2D crossover sits one step later than 1D is true only in the idealised count, and on real connectivity the 2D crossover is at n=4 as well. The 1D n=6 advantage rises 10.78x -> 15.36x. **This is the direction that could have gone against us and did not; had it gone the other way the headline would have needed revising, which is why it was checked before presenting.** Applies to gate counts only: it says nothing about whether either circuit's OUTPUT survives hardware noise, which would need QPU time and is not claimed here | deterministic (fixed transpile seed 2026, optimization_level=1, real backend coupling map) | 0 shots (compilation only) | `python evidence/scripts/aqc_routing.py ibm_marrakesh`; evidence/aqc_routing.json | 2026-08-14

R053 | **A high |chi| SURVIVAL can hide a badly wrong chi — demonstrated on the teammate `ibm_miami` data we already had, and it is a warning about our own reporting metric** | The `real_machine/` jobs (R031) were analysed but never used on a slide. Re-reading them for the complex chi rather than its magnitude turns up a case worth reporting. **FULL TABLE (single fixed-basis chi job per cell, 2000 shots): kingston/frozen: survival 0.808, |chi-exact| 0.1397, worst 6.4 sigma; kingston/2site: survival 0.595, |chi-exact| 0.2623, worst 12.4 sigma; kingston/4site: survival 0.587, |chi-exact| 0.3014, worst 13.4 sigma; miami/frozen: survival 0.738, |chi-exact| 0.2023, worst 8.5 sigma; miami/2site: survival 0.988, |chi-exact| 0.4847, worst 22.6 sigma; miami/4site: survival 0.300, |chi-exact| 0.5356, worst 17.9 sigma. THE ROW THAT MATTERS: ibm_miami / 2site reports |chi| survival 0.988 — which reads as a near-perfect run — while the COMPLEX chi is wrong by 0.485 on a quantity of magnitude 0.647, a phase error of 0.772 rad, and 22.6 sigma from exact.** The magnitude survived; the phase did not. `ibm_miami / 4site` shows the same pattern less extremely (survival 0.300, phase error 0.704 rad). **WHY THIS MATTERS BEYOND MIAMI:** 'survival' = |chi_hw| / |chi_exact| is the summary statistic used throughout this project's hardware rows (R008, R018, R023, R044). This is a measured example of it being actively misleading. **We checked our own headline rows: R023 and R044 both report the COMPLEX chi with per-quadrature sigma deviations alongside survival, so neither is affected — but that is a property of how they happen to be written, not of the metric.** Rule adopted: never report survival alone. **IT IS ALSO THE SAME LESSON AS R046, arrived at independently and from real data:** there, a phase-blind state-fidelity objective silently broke a Hadamard test; here, a phase-blind summary statistic silently flatters a hardware run. A Hadamard test is an interferometer, and any magnitude-only figure of merit is structurally blind to what it measures. **WHAT WE DO NOT CLAIM:** no diagnosis of why miami's phase drifted. Single run per model, a different IBM account/token, calibration state unknown to us, and one fixed basis per job — this is an observation about the metric, NOT a characterisation of the device, and miami is not presented as worse or better than kingston (the survival ordering between them is inconsistent across the three models: ratios 1.10, 0.60, 1.96). ARCHITECTURE NOTE (2026-08-14, see R031 update): miami is a Nighthawk-class square-lattice device, so this phase observation is from a different architecture than every other row in the project — added context, not a new claim; a single run still supports no architecture comparison | single run per cell, 2000 shots, no repeats; sigmas from the per-quadrature sem in the same analysis path as every other hardware row | reuses R031's shots, 0 new QPU time | evidence/real_machine_analysis.json; `python evidence/scripts/load_real_machine_results.py` | 2026-08-14

R054 | **THE AQC COMPILATION ADVANTAGE DOES NOT TRANSLATE TO HARDWARE AT n=6 — our own pre-registered prediction, falsified. Reported as the headline it is** | Three ways to build the SAME Hadamard test's controlled evolution, submitted together to `ibm_marrakesh` (job `d9uv5h50vrcc73boj8a0`, 18 circuits x 4000 shots, n=6 system qubits + ancilla, t = 0.3/0.6/0.9). Predictions were recorded BEFORE submission (R052 and the script's own output) precisely so this could fail publicly. **RESULT: exact: 8850 2q gates, predicted survival 3.60e-07, MEASURED 1.485, mean |chi err| 1.2627; trotter: 2119 2q gates, predicted survival 1.43e-02, MEASURED 0.033, mean |chi err| 0.8218; aqc: 576 2q gates, predicted survival 3.15e-01, MEASURED 0.026, mean |chi err| 0.8277. ALL THREE ARMS RETURNED NOISE. The 576-gate AQC arm — the one whose whole purpose was to be shallow enough to survive — came back at 0.026 survival against a predicted 0.315, i.e. 12x worse than predicted and indistinguishable from the 2119-gate Trotter arm (0.033). On mean |chi| error the Trotter arm is nominally closest to exact (0.822 vs AQC 0.828), but both are ~0.83 on a quantity of magnitude <=1, so neither measured anything. THE EXACT-BLOCK ARM IS A TRAP, AND IT IS THE MOST INSTRUCTIVE PART OF THE RUN.** Its 'survival' reads **1.485**, the best of the three — but that is unphysical: it measured chi = +0.8525+0.8580j / +0.8245+0.8555j / +0.8495+0.8325j at t = 0.3/0.6/0.9, i.e. |chi| ~ 1.21 against a bound of |chi| <= 1, essentially CONSTANT across all three times. Both quadratures are pinned near +0.85, which is the ancilla sitting in |0>: **T1 relaxation, not signal.** Diagnosis quantified: 8850 two-qubit gates x 68 ns median cz duration = 602 us against marrakesh's median T1 = 160 us (3.8 T1) and T2 = 74 us (8.1 T2). A magnitude-only survival metric scores this dead circuit as the BEST arm. **This is the third independent instance in this project of a magnitude hiding a broken phase — after R046 (a phase-blind AQC objective) and R053 (a phase-blind summary statistic on ibm_miami) — and the first where it appears in OUR OWN headline experiment. MECHANISM, PARTIALLY EXPLAINED AND HONESTLY SO:** the pre-submission model used two-qubit gate error only. Folding in ancilla dephasing (exp(-t/T2) = 0.59 for the AQC arm at 39 us) brings the prediction 0.315 -> 0.186, still ~7x above the measured 0.026. **We do not fully**

explain the residual factor and do not invent a cause; plausible contributors not separated here are routed layout edges worse than the median, crosstalk, and coherent error in a dense 7-qubit entangled ansatz. **CORRECTION (2026-08-14, C7 audit): the effective-error figure is a LOWER BOUND, not a measurement** — at 4,000 shots/quadrature the |chi| estimator's noise floor is sigma*sqrt(pi/2) ~ 0.020, and the measured 0.026-0.033 sits near it, so eps_eff >= ~6e-3 and the gap to visibility is >= 3x; A7 and the study notes now say 'bounded below'. The lesson we DO draw: a gate-count-based survival prediction is not adequate at this depth, and R042's calibrated model (which matched R023 at 15 gates) does not extrapolate to 576+. WHAT SURVIVES AND WHAT DIES. SURVIVES: every compilation claim — R046 (crossover n=4, 36x by n=7), R049 (2D), R051 (Hubbard), R052 (routing widens the gap) — these are transpiled gate counts and statevector fidelities, unaffected by any hardware outcome. DIES: any claim that the gate-count advantage becomes a* measured *hardware advantage at n=6 on today's devices. It does not, and the decks now say so. KINGSTON RETURNED (job* `d9uvhdob1g9c73a7vrl0`*, 2026-08-14) AND REPRODUCES THE FAILURE ON A SECOND DEVICE: exact: predicted 7.87e-07, MEASURED 1.191, mean |chi err| 1.0804; trotter: predicted 9.08e-03, MEASURED 0.016, mean |chi err| 0.8112; aqc: predicted 4.01e-01, MEASURED 0.033, mean |chi err| 0.8356. The AQC arm measured 0.033 against a predicted 0.401 (12x short, the same factor as marrakesh's 12x), Trotter 0.016, and the exact block again reports an unphysical apparent survival of 1.191 (chi = +0.6735+0.6595j / +0.6810+0.6760j / +0.6995+0.6680j, again near-constant and both quadratures pinned, now near +0.68 rather than +0.85 — consistent with kingston's longer coherence leaving the ancilla less completely relaxed). So the negative is not a one-device artefact: two independent QPUs, 144,000 shots total, same verdict — the compilation advantage does not reach hardware at n=6, and on both machines a magnitude-only metric would have crowned the deadest circuit** | single run per configuration, 4000 shots/circuit, no repeats; predictions recorded pre-submission | 72,000 shots | `python evidence/scripts/aqc_hardware_submit.py ibm_marrakesh`; `aqc_hardware_fetch.py`; evidence/aqc_hw_result.json; chart evidence/scripts/make_aqc_hw_chart.py | 2026-08-14

R055 | **The n=7 job, fully transpiled and costed — the arithmetic behind choosing n=6 for hardware** | Ran the unmodified submit script at n=7 against the real ibm_marrakesh target (dry run, no submission, no QPU time) after adding an `--n` flag. **ROUTED TWO-QUBIT COUNTS AT n=7: exact block 36,114 (vs 8,850 at n=6, ratio 4.08x, matching the ~4x claim in R054's footnote); Trotter r=2 2,549; AQC 702 (= 468 ideal x 1.50 routing inflation, identical inflation to n=6's 576/384).** Gate-error-only survival predictions: exact 3.08e-28, Trotter 6.04e-03, AQC 2.45e-01. AQC compile at n=7: state infidelity 5.1e-06..2.9e-04, pre-flight identity passed. **THE DECISION ARITHMETIC, all from measured quantities:** durations at 68 ns median cz — exact 2.46 ms = 15.4 T1 = 33.2 T2 (marrakesh T1 160 us, T2 74 us); AQC 47.7 us. Total per-shot circuit time (36,114+2,549+702) x 68 ns = 2.68 ms vs 785 us at n=6 — **the whole job costs 3.4x more quantum time, the exact control arm alone 4.1x**. Applying the MEASURED effective error from R054 (5.9-6.3e-3/gate) to the 702-gate AQC arm: predicted survival exp(-~6.1e-3 x 702) ~ 0.012-0.016, i.e. **half of n=6's already-failed 0.026-0.033**. Meanwhile the compilation RATIO improves 16,812/468 = 35.9x. **So n=7 doubles the ratio's headline, halves the measurable signal, and quadruples the bill — the quantitative form of 'relative advantage up, absolute feasibility down'.** Visibility threshold restated: signal ~0.3 at this depth needs effective error <= ln(1/0.3)/576 = 2.1e-3; measured is ~6e-3 | deterministic (transpile seed 2026, real backend target); statevector pre-flight 4e-16..8e-3 as at n=6 | 0 shots (dry run) | `python evidence/scripts/aqc_hardware_submit.py ibm_marrakesh --n 7 --dry-run`; console capture /tmp promoted to evidence/n7_dryrun_console.txt | 2026-08-14

R056 | **The AQC gate-scaling trend EXTENDED from n=7 to n=10 — same script, same method, unmodified — and the exponential separation keeps widening exactly as the 4^(n+1)-vs-linear argument predicts** | R046 measured the crossover at n=4 and reported 35.9x by n=7; the claim on-slide was that the win keeps growing but that was extrapolation past n=7, not measurement. `aqc_scaling.py`'s FIELDS tuple only covered n<=7, so 3 more entries were appended (identical values to `skqd_scaling.py`'s existing FIELDS continuation, reused rather than invented, so the two scripts' Hamiltonian families stay consistent) and the unmodified script run at n=8, 9, 10. **RESULT (controlled 2-qubit gate counts, ideal basis-gate count, no routing): n=8: exact 67,756 vs AQC 552 = 122.75x, state infidelity 2.90e-04, |chi err| after phase fix 0.0083; n=9: exact 272,044 vs AQC 636 = 427.74x, infidelity 2.90e-04, |chi err| 0.0083; n=10: exact 1,090,220 vs AQC 720 = 1514.19x, infidelity**

2.81e-04, |chi err| 0.0082. Full trend, n=3..10: 0.38x, 1.09x, 3.35x, 10.78x, 35.92x, 122.75x, 427.74x, 1514.19x — monotone, and the ratio between consecutive points (~2.9-3.6x per qubit) tracks the exact block's 4x-per-qubit growth divided by AQC's near-flat cost, exactly the structural argument R046 made before any of these numbers existed. AQC state infidelity and the phase-fixed chi error are both FLAT across n=8..10 (2.8-2.9e-4 and 0.0082-0.0083 respectively) — the compressor is not straining as the system grows, at least up to 10 system qubits (11 with the ancilla). Exact-block unitary synthesis time grew 4.6s (n=8) -> 26.7s (n=9) -> 125.4s (n=10), consistent with the same 4^n scaling that makes the exact path the loser; this is also why the sweep was stopped at n=10 rather than pushed further — n=11's synthesis alone was projected past 500s and adds no new qualitative information the n=3..10 trend does not already show | deterministic (transpile seed 0, L-BFGS-B maxiter as in R046, single run per n — gate counts and infidelities have no shot noise, matching every other row in this family) | 0 shots (classical compilation study) | `python evidence/scripts/aqc_scaling.py 8`; `... 9`; `... 10`; evidence/aqc_scaling.json (now holds the full merged n=3..10 trend) | 2026-08-14

R057 | **R047's SKQD Delta/J boundary CONFIRMED under 20-seed bootstrap — the monotone trend survives, with two honest calibration corrections against the single-seed point estimates** | R047 explicitly flagged its own gap: the Delta/J control sweep (n=12 sector, dim 924) was "single-seed (RNG 2026), no bootstrap." House rule wants >=10 seeds and a bootstrap 95% CI on stats claims. The sector Hamiltonian and its eigendecomposition are deterministic (no RNG) and were reused unchanged per Delta/J point; only the sampling step (which configurations the sampler observes from the time-evolved Neel reference) is stochastic, so that step was repeated across 20 independent seeds and bootstrapped (10,000 resamples) at each of the same 5 Delta/J values as R047. **RESULT (mean dim needed for 1% error / 924, 95% CI in brackets): Delta/J=0.38: 84.4% [81.2%, 86.6%]; 1.00: 48.7% [45.5%, 53.0%]; 2.00: 18.9% [17.9%, 20.0%]; 5.00: 2.7% [2.7%, 2.7%] (zero variance -- always hits the smallest grid point M=25); 10.00: 2.7% [2.7%, 2.7%] (same). The monotone-decreasing trend across Delta/J is CONFIRMED at 20 seeds (verified programmatically, not by eye). TWO CALIBRATION NOTES against R047's single-seed values:** at Delta/J=1.00, R047 reported 43.3%, just below the 20-seed CI's lower bound (45.5%) -- the single draw was mildly optimistic; at Delta/J=2.00, R047 reported 16.2%, also just below the CI (17.9%) -- same direction, same size of effect. At Delta/J=0.38 (our benchmark's own ratio) and at the two localised points (5.00, 10.00), R047's single-seed numbers sit inside or exactly on the 20-seed estimate. **CONCLUSION: R047's qualitative finding -- SKQD needs a shrinking fraction of the sector as Delta/J grows, and needs the MAJORITY of a large delocalised sector at our benchmark's own ratio -- is now a statistically supported trend, not a single-seed anecdote; the only correction is that the delocalised-regime numbers were slightly better than reality by 2-3 percentage points** | 20 seeds per Delta/J point, bootstrap 10,000 resamples, 95% CI reported (satisfies the >=10-seed house rule this row exists to close) | 0 shots (classical; sampling simulated from exact statevectors, same protocol as R047) | `python evidence/scripts/skqd_boundary_bootstrap.py`; evidence/skqd_boundary_bootstrap.json | 2026-08-14

R058 | **Hardware error-rate trend: a shots sweep on the R054 AQC arm — does the survival floor track shot-noise, or is there a real error floor above it? Evidence points to the latter, without claiming causality from single runs** | R054/A7 bound the AQC arm's effective error as eps_eff >= ~6e-3/gate, flagged by the C7 audit as a LOWER bound because measured |chi| sits near the estimator's own counting-noise floor sigma*sqrt(pi/2) at 4,000 shots. If that bound is mostly counting-noise artefact, survival should fall roughly as 1/sqrt(shots) as shots increase; if it reflects a real device-error floor, survival should plateau. Submitted two more jobs to `ibm_marrakesh` at the SAME n=6 model, SAME 3 times, SAME 576-gate AQC arm as R054 (job `d9v95ufo3ppc73ak01o0`, 2000 shots; job `d9v9640b1g9c73a8b660`, 8000 shots), bracketing R054's original 4000-shot job (`d9uv5h50vrcc73boj8a0`) on both sides. RESULT (AQC arm, mean survival over the 3 times): 2000 shots -> 0.0659; 4000 shots (R054) -> 0.0261; 8000 shots -> 0.0244. Predicted pure counting-noise floor at each shot count: 2000 -> 0.0280; 4000 -> 0.0198; 8000 -> 0.0140. Measured survival sits 1.3-2.3x above the floor at every point, not a fixed ratio, so pure shot noise alone does not explain it. THE INFORMATIVE COMPARISON is 4000 -> 8000: doubling shots should shrink the floor by sqrt(2) (0.0198 -> 0.0140, a 29% drop), but measured survival barely moved (0.0261 -> 0.0244, a 7% drop) -- consistent with a real error floor sitting above the counting-noise bias rather than the bound simply needing more shots to tighten away. Recomputing eps_eff = -ln(survival)/576 at each point: 2000 shots -> 4.7e-3;

4000 shots -> 6.3e-3 (R054's number); 8000 shots -> 6.4e-3 -- the bound RISES with shots and stabilises between 4000 and 8000, which is the expected behaviour if ~6-6.5e-3 is close to the true effective error rather than an artefact of too few shots at 4000. The exact arm reproduces R054's unphysical-survival pattern at both new shot counts (mean survival 1.40 at 2000, 1.42 at 8000; both ~1.2 mean |chi err|) -- the T1-relaxation floor is not a 4000-shot artefact either. WHAT THIS DOES NOT ESTABLISH: single run per shot count, submitted as three separate jobs at different times on the same backend, so device drift between submissions is a fully confounded alternative explanation and no causal claim is made from shot count alone; house rule on single-run causal claims applies. A CLOBBER BUG FOUND AND FIXED WHILE SUBMITTING THIS ROW:* `aqc_hardware_submit.py`'s job-record key was `backend+n` only, so a shots-only rerun at the default n=6 would have silently overwritten R054's original marrakesh job entry; fixed to key on `backend+n+shots`. A second instance of the same clobber class was then caught in `aqc_hardware_fetch.py`, which overwrote the entire `evidence/aqc_hw_result.json` (losing the already-fetched R054 marrakesh/kingston analysis) the moment it was pointed at a single new backend key; recovered from git history and the fetch script fixed to merge | single run per shot count (2000/4000/8000), no repeats at fixed shots; the trend itself is 3 points, not independently bootstrapped | 20,000 new shots (18 circuits x 2000 + 18 x 8000); reuses R054's original 72,000 shots for the 4000-shot point | `python evidence/scripts/aqc_hardware_submit.py ibm_marrakesh --shots 2000`; `... --shots 8000`; `aqc_hardware_fetch.py ibm_marrakesh_s2000`; `... ibm_marrakesh_s8000`; evidence/aqc_hw_job.json; evidence/aqc_hw_result.json | 2026-08-14

R059 | **Track B on hardware (p.20/21, R044): the time window EXTENDED from t<=2.7 to t<=5.4 on ibm_marrakesh — the echo's lead holds throughout, and the free error bar stays bounded rather than growing with t** | R044 measured chi_AB(t), the per-observable W/U profile, and the free error bar “_W - ‘_U at only 4 times, t = 0.0/0.9/1.8/2.7 (about one Trotter step's worth of angle). Whether the echo's advantage or the charge-conservation error bar are artefacts of that short window was unmeasured. Submitted a NEW job to `ibm_marrakesh` (job `d9v9djf2sl0c73bljm8g`, 203 circuits x 1000 shots = 203,000 shots) with TIMES extended to (0.0, 0.9, 1.8, 2.7, 3.6, 4.5, 5.4) -- R044's 4 points plus 3 more at the same 0.9 step, doubling the window. **Deliberately kept as a SEPARATE script and SEPARATE evidence files** (`track_b_hardware_time_extend.py`, `evidence/track_b_hw_jobs_ext.json/track_b_hw_result_ext.json`): the existing pipeline's chart generator (`make_trackb_charts.py`) assumes one global TIMES/exact_reference and a single-job result list, and retrofitting that for a second, longer-TIMES job risked exactly the clobber-bug class already fixed twice today in the AQC scripts, so this keeps R044's original evidence and the current p.20/21 figures completely untouched. **PHYSICS NOTE BEFORE THE NUMBERS: circuit depth here does NOT grow with t** -- only the rotation angle inside `build_controlled_evolution` changes, so all 7 points ran the identical 136-two-qubit-gate, depth-540 circuit (confirmed: median_2q 136, matching R044 exactly). Any trend across t therefore reflects the algorithm/physics, not accumulating circuit-depth decoherence. **RESULT 1 -- the echo's lead holds over the full window:** Loschmidt echo P0 measured/exact at the 3 NEW points: t=3.6: 0.2760/0.2389; t=4.5: 0.1950/0.1984; t=5.4: 0.2590/0.2673 -- mean |echo error| across all 7 points (old + new) is 0.0166, worst 0.0371 at t=3.6, essentially unchanged from R044's ~0.04-scale error at t<=2.7. **RESULT 2 -- the Track-B ancilla arm (chi_AB) does not catch up:** mean |chi_AB error| across all 7 points is 0.234, i.e. the echo remains ~14x more accurate over the extended window, consistent with (not better or worse than) R044's original 27x-on-the-full-scoreboard finding. **RESULT 3, the one most relevant to p.21 -- the free error bar STAYS BOUNDED, not growing, as t increases:** measured ‘_W - ‘_U at the 7 points: +0.009, -0.053, -0.209, -0.064, -0.137, -0.024, -0.199 (exact identity: 0 at every t). Mean -0.097, std 0.086, range [-0.209, +0.009] -- no monotonic drift with t (the largest-magnitude point, t=1.8, is bracketed by smaller ones on both sides at t=1.35-ish and t=2.7), consistent with the diagnostic remaining a bounded, reference-free error indicator across a 2x-longer window rather than one that degrades as the experiment runs longer. **AN HONEST DISCREPANCY, not resolved here: these 7 deviations are systematically SMALLER than R044's original marrakesh numbers at the 3 overlapping times** (R044: -0.044/-0.393/-0.251/-0.335 at t=0/0.9/1.8/2.7; this row: +0.009/-0.053/-0.209/-0.064 at the same t's) even though the chi_AB and echo numbers at those same overlapping times are close to R044's (e.g. echo 0.9620 here vs 0.9650 in R044 at t=0.9 -- a consistency check that this is measuring the same physics). **No cause is claimed**: this is a different job submitted at a different time on the same backend, so device calibration drift between submissions is a live, fully confounded alternative explanation, and single run per point (no repeats) means neither job's numbers get to overrule the other -- both are reported | single run per time point, no repeats, no bootstrap (hardware, exploratory, same discipline as R044) | 203,000 shots (7 times x (9 bases x 3 arm-A/B circuits + 2 baseline circuits) x 1000) | `python evidence/scripts/track_b_hardware_time_extend.py ibm_marrakesh`; `track_b_hardware_time_extend_fetch.py`; evidence/track_b_hw_jobs_ext.json; evidence/track_b_hw_result_ext.json | 2026-08-14’’’”

R060 | **"Pure AQC" -- no Hadamard test, no shadow -- does NOT hit the phase trap at all: it is safe for its textbook use case (state prep + direct observable measurement), and the trap is specific to interferometric use** | R046 found AQC's compiled controlled-U is wrong by |chi err|~1.3 when fed into an interferometric Hadamard test. A judge asked the DIFFERENT question: did we compare against "pure AQC" -- no ancilla, no controlled-U, no interference circuit of any kind, just AQC used the way the compiling literature normally uses it, to compress a state-preparation circuit and measure ordinary observables directly on the result? **THE ALGEBRAIC ARGUMENT, stated before measuring:** chi = is NOT an observable expectation value in any single state -- it is an interference term between |psi> and |U psi>, and extracting it PHYSICALLY requires an ancilla-based circuit (Hadamard test or an informationally equivalent one: Loschmidt echo, Hilbert-Schmidt test). There is no way to "just measure" chi. By contrast, _W = needs no ancilla -- ordinary Pauli sampling on the AQC-prepared state suffices. And _W is EXACTLY invariant under the phase ambiguity that breaks chi: if W|psi> = e^{i theta} U|psi> (R046's own finding), then _W = = _U for ANY theta and ANY Hermitian O -- so pure AQC state-prep should not trip the phase trap at all. **VERIFIED, not just derived, at n=3/5/7 (subset of R046's trend), reusing the SAME AQC-compiled W as R046 (chi errors reproduced as a same-n cross-check: 1.3498/1.2912/1.3129 vs R046's 1.350/1.291/1.313 -- confirms this is the identical compiled circuit, not a different run):** direct-observable errors on “, , (energy and system observables, no ancilla): worst error across all three observables and all three n is **0.0182 (n=3), 0.0144 (n=5), 0.0147 (n=7)** -- two orders of magnitude smaller than chi's ~1.3 error, and consistent with AQC's ordinary state infidelity (2-3e-4), not with any phase pathology. **EXPLICIT PHASE-PERTURBATION CHECK (not just the algebra):** multiplying the AQC-prepared state by an ARBITRARY extra global phase e^{i theta_extra} (random theta_extra per n) changes every measured by <=2.78e-16 -- floating-point zero, confirming the invariance is exact, not merely small. **CONCLUSION FOR THE JUDGE'S QUESTION: yes, we effectively compared both regimes. "Pure AQC" (no Hadamard test, no shadow) is SAFE and accurate for its textbook use case -- direct state-prep and observable estimation -- because that use case structurally cannot see the phase error at all. The trap is not a defect in AQC; it is a hazard specific to embedding AQC inside an interferometric protocol (ours, or any other), which is exactly the scenario our project's whole verification apparatus (Track B, the shadow ensemble, the phase-fix gate) exists to catch and correct.** This sharpens rather than weakens the project's core claim: the phase trap is real and would have silently broken our result, but it says nothing bad about AQC used as intended -- our contribution is identifying the one place (interferometric verification) where AQC's own certificate is the wrong metric, not a general critique of the compiler | deterministic (fixed transpile seed, optimization_level=1; state-vector calculation, no shot noise); phase-perturbation check exact (0 to 1e-15, floating-point) | 0 shots (classical statevector study, 0 QPU time) | `python evidence/scripts/aqc_direct_observable.py 3 5 7`; evidence/aqc_direct_observable.json | 2026-08-14”